Reducing, Holding and Expanding the Size of the State: A Modular Stochastic Framework for Evaluating Fiscal Reform Scenarios in Brazil, with an Application to the 2026 Presidential Programmes

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26th October 2026

Abstract

This paper documents a modular, stochastic simulation framework evaluating what happens, over five- to twenty-year horizons, when Brazilian general-government primary spending is reduced, held against demographic ageing, or expanded, relative to a status-quo baseline. Nine chained blocks — a demographic fiscal baseline, growth and debt accounting, a household microsimulation on the 2025 PNAD Contínua, social outcomes, long-run human- and physical-capital feedback, an external shock block, and an explicit governance layer — are propagated through a thousand-draw Monte Carlo that reports paired differences against the baseline. The same machinery, without re-specifying a single equation, translates the two 2026 run-off candidates' filed fiscal programmes into comparable paths. We report the model's own historical back-test honestly: differences against the baseline pass a stated tolerance, but the absolute public-debt level does not, and we argue a stochastic stock-flow-adjustment term is a more honest response to that failure than a fitted correction. In direct response we add a set of clearly labelled, off-by-default diagnostics — breach probabilities, a fiscal-reaction-function check, a nonlinear risk-premium variant, and six further extensions built in response to an independent review — without ever presenting an absolute debt level as validated; one of those extensions reverses the sign of the paper's own headline distributional finding once interest income is allocated the way it is actually held. The framework's principal contribution is not a forecast but a disciplined, falsifiable, and fully traceable accounting of what a given fiscal composition and size imply, and of how much of that implication is assumption rather than estimate.

Keywords: fiscal consolidation; debt sustainability; Monte Carlo microsimulation; state size; Brazil; corruption and governance; fiscal multipliers.

JEL classification: H62 (deficit; surplus); H63 (debt; debt management); H50 (national government expenditures and related policies); C63 (computational techniques; simulation modelling); I38 (government policy; welfare and poverty programmes); D31 (personal income, wealth and their distributions).

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1 Introduction

A perennial question in the political economy of public finance is how much the state should tax and spend, and how the answer changes an economy's growth, its public debt, and the distribution of income and health across its population. Brazil poses this question with unusual sharpness. General-government primary spending reached 39.8% of GDP in 2025, population ageing is projected to raise the age-related share of that spending by a further 3.3 percentage points of GDP (pp GDP) over the subsequent decade even with no policy change (IBGE, 2024; Secretaria do Tesouro Nacional, 2025), and gross general-government debt stood at 78.6% of GDP at the close of 2025 (). The 2026 presidential run-off has, in turn, placed two sharply different fiscal philosophies before voters: a constitutional spending-growth rule tying real federal expenditure growth to a fraction of real revenue growth, and a debt-stabilising primary-balance rule that moves the government's balance linearly towards the level implied by the debt dynamics identity.

This paper documents a model built to make that debate tractable without pretending to resolve it. We did not set out to forecast Brazil's fiscal future, nor to rank the candidates. We set out to build an instrument that takes a stated change in the size and composition of the state — expressed, deliberately, in the same units a finance ministry would use, percentage points of GDP — and propagates it consistently through growth, debt, household incomes, health outcomes, long-run human- and physical-capital accumulation, the external sector, and an explicit account of leakage and corruption risk, while reporting at every step how much of the result is a sourced estimate, a defensible modelling choice, or an acknowledged gap. Every numerical parameter in the model is traceable to a single registry file with its source, vintage and licence; where no credible estimate exists the model raises an error rather than silently substituting a plausible number. This discipline is, we think, the paper's main methodological claim: a scenario model's usefulness depends less on the sophistication of any one equation than on the auditability of the whole chain and the honesty with which it reports its own failures.

The remainder of the paper proceeds as follows. Section 2 places the model within eleven strands of literature, from the broad composition-of-consolidation and fiscal-multiplier debates through stochastic debt-sustainability analysis, debt-threshold and interest-growth-differential dynamics, the distributional and social consequences of adjustment, microsimulation and fiscal-incidence methodology, sovereign risk premia and fiscal fatigue, governance and corruption, fiscal-rule design, and the stock-flow-adjustment literature that turns out to anticipate this paper's own central finding. Section 3 sketches the Brazilian fiscal and demographic context the model takes as given. Section 4 describes the modular architecture and the nine chained blocks. Section 5 defines the baseline, reduction and expansion scenarios and explains how the two 2026 candidate programmes are mapped onto the same machinery. Section 6 describes the Monte Carlo design and the uncertainty it does and does not capture. Section 7 reports the model's historical back-test honestly, including the dimension on which it fails, and summarises the diagnostic extensions, given in full in Appendix B, built in direct response to that failure. Section 8 summarises the headline, gate-cleared results. Section 9 discusses limitations and the appropriate use of the model, and Section 10 concludes.

To anchor what follows in numbers before the mechanics behind them are introduced: under the model's own status-quo baseline, debt/GDP rises from 78.6% in 2025 to a central 145.5% by 2035, driven mainly by the interest-growth differential rather than discretionary policy (Section 8); against that baseline, a central, sustained 4 pp-of-GDP consolidation lowers ten-year debt by a little under 24 pp of GDP at a median GDP cost of under one percentage point under transitory multipliers; and the model's own historical back-test fails at the level of absolute debt, though not at the level of paired scenario differences (Section 7), which is

why every comparison in this paper is reported as a difference against the baseline, never as a validated level. A later diagnostic (Appendix C, Section 9) finds that one of the paper's own headline distributional results reverses sign once a channel the published Monte Carlo does not model is added — a finding we report in full rather than soften, for the same reason we report the back-test failure in full.

2 Related literature

Across these eleven strands, one gap recurs: existing debt-sustainability, multiplier and distributional literatures are each well developed on their own, but rarely chained into a single, auditable instrument that propagates one stated policy change through growth, debt, household incomes and social outcomes together, and checks the result against the country's own history before reporting it. That gap is what this paper tries to fill. This section positions the model against the eleven strands; Appendix A gives the full discussion, with every supporting citation, theme by theme. The summary here keeps only the most load-bearing references in each strand.

On the *composition of fiscal consolidation*, Alesina and Ardagna (2010); Alesina et al. (2015) find spending-based adjustment plans produce milder output losses than tax-based ones, a finding Guajardo (Leigh and Pescatori) contest using narrative identification that rejects "expansionary austerity" more broadly. We take no position: the model reports whatever composition the user specifies, using Brazil-specific rather than imported multipliers. On *multiplier size and state-dependence*, Auerbach and Gorodnichenko (2012) and Ilzetzki (Mendoza and Végh) are the central references, the latter showing multipliers fall with debt, trade openness and exchange-rate flexibility in ways that place an economy such as Brazil's towards the lower end of the range; Pires (2014) and Resende and Pires (2021) provide the Brazil-specific estimates this model's category-specific multipliers are checked against.

On *stochastic debt-sustainability analysis*, the model's direct methodological ancestor is Celasun (Debrun and Ostry)'s fan-chart approach, which Garcia and Rigobon (2004) and Karam and Hostland (2005, 2006) apply to Brazilian and emerging-market debt specifically; Levy Yeyati and Sturzenegger (2023) offer a more recent balance-sheet reformulation of the same question, and Prasad et al. (2019); Lian (Presbitero and Wiriadinata) point towards the conditional-quantile "debt-at-risk" framing that a resampled Monte Carlo such as this one approximates but does not formally implement (a limitation Section 9 returns to). On *debt thresholds and the interest-growth differential*, Reinhart and Rogoff (2010) and its correction by Herndon (Ash and Pollin) are the cautionary precedent against single-threshold summaries, reinforced by Égert (2015) and Panizza and Presbitero (2014); Blanchard (2019)'s finding that $r < g$ has historically been closer to the norm than the exception directly motivates the paper's own result that the interest-growth differential, not the primary balance, dominates Brazil's projected debt path.

On the *distributional and social consequences of adjustment*, Ball et al. (2013); Woo (Bova) establish that consolidation raises the Gini coefficient, more so when spending-based; Lago-Peñas (Cadaval-Sampedro and Herrero-Alcalde) add a political-economy dimension, showing voters' tolerance for austerity depends on its financing instrument and their own ideology. Szklo (Clarke and Rocha) and Rasella et al. (2013) are the Brazilian evidence behind the health-spending and cash-transfer channels respectively, read against Filmer and Pritchett (1999)'s scepticism that cross-country health spending alone explains mortality. On *microsimulation methodology*, Bourguignon and Spadaro (2006) sets out the comparative-statics case this paper's own household block follows; Higgins and Pereira (2014) apply the Commitment to Equity framework to Brazil directly, and Balladares and García-Miralles (2026) is a recent

demonstration, in this journal, of exactly this style of tax-record microsimulation applied to a live policy question.

On *sovereign risk premia and fiscal fatigue*, Baldacci and Kumar (2010) anchors this model's linear premium coefficient, while Ghosh et al. (2013) and Bi (2012) provide the nonlinear, fiscal-fatigue theory behind the diagnostic premium variant of Appendix B. On *governance and corruption*, Ferraz (Finan and Moreira); Avis (Ferraz and Finan) supply the Brazilian audit evidence calibrating this model's leakage share; Mauro (1995); Olken (2007) are the foundational cross-country references, and Hanousek and Kočenda (2011) link corruption directly to fiscal performance, debt and deficits, across the new EU member states, a finding structurally close to this paper's own governance layer. On *fiscal rules and the political economy of debt*, Kopits and Symansky (1998) provide the taxonomy within which this paper situates the two 2026 candidate programmes as, respectively, an expenditure rule and a balance rule, without ranking them; Feld and Kirchgässner (2008) and López-Villavicencio and Zoumenou (2025) are the closest real-world comparators.

Finally, and most consequentially for this paper's own validation result, the *stock-flow-adjustment* literature independently documents exactly the phenomenon behind this model's Tier 2 back-test failure (Section 7): Weber (2012) and Jaramillo (Mulas-Granados and Kimani) show such adjustments are a large, persistent, poorly-explained source of debt accumulation worldwide, and that large adjustments predict debt subsequently failing to decline, while Mackenzie and Stella (1996) explains why they occur at all (quasi-fiscal operations of public financial institutions, a description that fits BNDES's role in Brazil's own episode directly). Gobetti and Orair (2017) is the closest peer-reviewed reconstruction of the Brazilian episode specifically. Appendix A discusses all eleven strands, and every citation above, at full length.

3 Fiscal and demographic context

Brazilian general-government primary spending was 39.76% of GDP in the four quarters to 2025, against primary revenue of 38.97% of GDP on the same General Finance Statistics Manual 2014 (GFSM) basis, implying a primary deficit of 0.79% of GDP (Secretaria do Tesouro Nacional, 2025). Gross general-government debt stood at 78.64% of GDP in December 2025 (). Of primary spending, old-age and survivors' pensions (RGPS and, by proxy, RPPS) accounted for 11.65% of GDP, health for 5.03% and education for 5.10%, using the 2024 COFOG functional classification carried forward one year (Secretaria do Tesouro Nacional, COFOG). Brazil's population is ageing: using National Transfer Accounts age-profiles for 2003 and 2018 (National Transfer Accounts, Brazil) combined with the IBGE 2024 population revision (IBGE, 2024), age-related spending alone is projected to rise by 3.3 pp of GDP over ten years and 7.3 pp over twenty, relative to a constant-spending-share counterfactual, with no change in policy or generosity.

This is the backdrop against which the 2026 presidential run-off programmes are written. The incumbent's programme is built around the existing constitutional spending-growth rule introduced by Complementary Law 200/2023 ("arcabouço fiscal"), which ties real federal primary-spending growth to 70% of real revenue growth, bounded between 0.6% and 2.5% a year. The challenger's programme proposes an explicit debt-stabilising primary-balance target, moving the federal government's primary balance over an unspecified number of years towards the level that would stabilise, rather than reduce, the debt ratio at its then-current level, financed partly through privatisation of state-owned enterprises. Section 5 explains how the model turns each rule into a comparable annual fiscal path without altering a single underlying equation.

4 Model architecture

The model is organised as nine chained blocks, each with a narrow, tested public interface, so that later blocks consume only the outputs — never the internal state — of earlier ones. Every coefficient, elasticity, stock and share used anywhere in the chain is declared once, with its source, vintage and a verification flag, in a single parameter registry; a missing or unverifiable input raises an explicit error rather than being silently defaulted. Figure 1 sketches the dependency structure; we describe each block in turn.

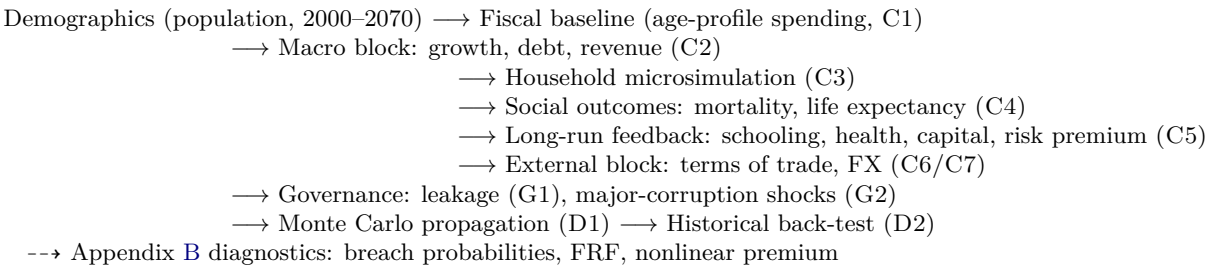


Figure 1: The model's chained block structure. Each solid arrow is a one-directional, tested data interface; no downstream block mutates an upstream one. The dashed arrow marks the Appendix B diagnostics, which branch off the back-test gate, run only against S0, and never feed back into the scenario chain above.

4.1 Demographic and age-profile fiscal baseline

The baseline block projects Brazilian general-government primary spending under two polar rules: a status-quo rule (S0) in which pension and health spending drift with the population's age structure while every other category is held at its 2025 share of GDP, and a constant-size rule (S1) in which total primary spending is held at today's share of GDP, so that ageing-driven cost growth in pensions and health must be offset elsewhere. Age-specific per-capita transfer profiles for education, health and old-age/survivors pensions come from Brazilian National Transfer Accounts for 2003 and 2018 (National Transfer Accounts, Brazil); because the two vintages straddle the 2019 pension reform, the model evaluates both, together with three indexation rules (pure per-capita cost growth, pure GDP-per-working-age-person growth, and an equal-weighted blend), and reports the minimum and maximum of six resulting combinations as low/high bounds, with the 2018 profile at equal weight as the central case. The resulting S0–S1 drift is 1.6/3.3/5.2/7.3 pp of GDP at five/ten/fifteen/twenty years (central estimates).

4.2 Macroeconomic block: growth, debt and revenue

The macro block converts an annual primary-spending path into output, debt and employment paths using a Cobb–Douglas production function,

$$Y_t = A_t K_t^{\alpha} (h_t L_t)^{1-\alpha}, \tag{1}$$

with capital accumulated endogenously, $K_t = (1-\delta)K_{t-1} + i_{t-1}Y_{t-1}$, rather than assumed to grow at a fixed rate. Capital share α, depreciation δ, the initial capital–output ratio and the investment rate are taken from Penn World Table 10.01 Brazil observations at constant 2017 prices (Feenstra, Inklaar and Timmer); total factor productivity and human-capital growth are drawn from three overlapping sub-period compound annual growth rates (2000–2019, 2004–2013 and 2010–2019) to produce low/central/high bounds, since no single historical window is

obviously the right one to extrapolate. Labour is population aged 15–64 at a fixed employment rate calibrated to its 2019 value; this omits any projected change in labour-force participation or the unemployment rate beyond an Okun-coefficient adjustment for demand-driven deviations.

Fiscal reductions or expansions below or above the S0 path are translated into output effects through Brazil-specific accumulated fiscal multipliers by spending category — public investment, social benefits, subsidies, and other current spending — drawn from Sanches (2026) — at the time of writing a working paper under review at *Estudos Econômicos* rather than yet formally published, a status we flag given how much of this block's calibration rests on it — and reported as both a transitory impulse-response and a persistent, permanently-lower-output variant, since the wider multiplier literature (Section A.2) does not agree on which horizon is appropriate — Auerbach and Gorodnichenko (2012) and Ramey and Zubairy (2018) reach materially different conclusions on state-dependence even within a single, advanced-economy sample — and the two horizons differ by an order of magnitude in their twenty-year debt consequences. Public debt accumulates through the standard identity

$$d_t = \frac{1 + i_t}{1 + g_t} d_{t-1} - pb_t + a_t, \tag{2}$$

where d_t is the debt-to-GDP ratio, i_t the effective nominal interest rate, g_t nominal GDP growth, pb_t the primary balance and a_t a residual debt adjustment (used, for instance, for privatisation proceeds or the stock-flow-adjustment noise term of Section 7). The effective real interest rate is estimated from the Fisher relation applied to five years (2021–2025) of observed interest payable over end-year gross debt and the contemporaneous GDP deflator, giving a historical range of −2.04% to 7.20% and a median of 5.05%, reinflated at each projection corner by the corner's own deflator assumption.

With no policy change, this block projects a central ten-year debt ratio of 145.5% of GDP by 2035 and a central ten-year potential growth rate of 0.94%, against an IMF benchmark of 2.5% (IMF, 2026); the gap is traced in Section 8 to a markedly weaker total-factor-productivity trend in the Penn World Table data than in IMF projections, and should be read as a statement about which historical window one chooses to extrapolate, not as an independent forecast superior to the IMF's. Revenue is held at a constant share of GDP under a unit elasticity assumption for the deterministic corners, with the Monte Carlo sampling a long-run elasticity of 0.92–0.98 from Brazil's independent fiscal institution (Instituição Fiscal Independente, 2021) for the stochastic runs; a scenario may additionally specify an explicit tax change, applied through the same Sanches et al. accumulated-multiplier machinery used for spending.

4.3 Household microsimulation

The distributional block is a static arithmetic microsimulation, in the comparative-statics tradition set out by Bourguignon and Spadaro (2006) and applied to Brazil specifically within the Commitment to Equity framework by Higgins and Pereira (2014) (Section A.6), of the 2025 PNAD Contínua (first-visit) cross-section (IBGE, PNAD Contínua), taking as inputs one macro-block scenario-year's GDP-level gap, unemployment change, package-specific spending gaps and revenue change, and returning the Gini coefficient, two absolute poverty headcounts (at the statutory Bolsa Família eligibility line and at half the minimum wage), and survey top 1% and top 10% income shares. Pension changes are allocated either in strict proportion to each benefit or, in an indexation-floor variant, in a way that protects benefits at or below the minimum wage; labour-income changes move either purely with the aggregate GDP level or through an explicit reallocation of simulated employment status consistent with the macro block's unemployment change; tax or transfer changes are allocated by household income, by

a consumption-like proxy, or — when a scenario specifies the taxing instrument explicitly — by an instrument-specific incidence base (top-income, payroll, consumption, or tax-expenditure beneficiaries). No behavioural, price, labour-supply or general-equilibrium response beyond the macro block's own GDP and employment effects is modelled; this is a comparative statics exercise applied to a single cross-section, not a dynamic microsimulation. A validated no-change run reproduces the official PNAD Gini (0.5112) and population totals to within source-data precision.

4.4 Social outcomes

Health spending per capita, net of an explicit allowance for lost services under an efficiency-savings package, is linked to infant mortality through an elasticity of −0.126 (range −0.071 to −0.206) estimated by Szklo (Clarke and Rocha) around Brazil's EC 29/2000 health-spending mandate, and to life expectancy through a single central elasticity of 0.04 from the cross-country meta-regression of Gallet and Doucouliagos (2017); both elasticities are applied symmetrically to spending increases and cuts, a simplifying choice flagged explicitly in the documentation since the underlying Brazilian estimate found some asymmetric evidence that mortality responds more to increases than to cuts. Elasticities build up linearly to their full effect over ten years, consistent with the estimation horizon of the source study, and are held flat thereafter. No schooling, physician-access or composite Human Development Index outcome is produced, because no credibly sourced elasticity for those outcomes against Brazilian fiscal spending was identified; those cells are explicitly marked as not modelled rather than left blank or imputed.

4.5 Long-run feedback channels

A separate block adds four long-run channels through which a scenario's spending path feeds back into output and debt beyond the horizon of a conventional fiscal multiplier: a schooling-to-human-capital channel, in which each school-age birth cohort's exposure to a change in education spending shifts its adult human capital according to the Penn World Table's Mincerian schooling-return function (Feenstra et al., Human Capital in PWT), combined with a Brazilian spending-to-schooling elasticity of 0.35 from Litschig and Morrison (2013); Litschig (2012); a health-to-output channel sampled uniformly between the near-zero estimate of Acemoglu and Johnson (2007) and the sizeable estimate of Bloom (Canning and Sevilla), since the two cannot be reconciled from the data available here; a public-capital channel using the output elasticity of Bom and Ligthart (2014), which remains switched off in the published runs for want of a Brazilian public-capital stock and depreciation input from the IMF Investment and Capital Stock Dataset; and a sovereign-risk-premium channel, in which a scenario's debt deviation from its own base year adds a premium of 2–7 basis points per percentage point of GDP (central 5bp, Baldacci and Kumar, 2010) to its effective real interest rate, capped at a 50-point deviation, resolved recursively year by year; this linear, capped specification is a deliberately modest reduced-form approximation to the richer nonlinear, fiscal-fatigue-style premium theorised by Ghosh et al. (2013) and Bi (2012) (Section A.7), an alternative, explicitly off-by-default specification of which is reported as a diagnostic in Appendix B. Each channel's contribution to the headline GDP and debt paths is reported separately, computed as a full re-run with that channel switched off, so that interaction effects are captured rather than assumed additive. An inequality-to-growth channel following Ostry (Berg and Tsangarides) is specified but left inactive pending a wired path for the Gini change produced by the household microsimulation.[1]

[1]Two separate obstacles, not one. First, a genuine circularity: the required Gini path comes from the household microsimulation (C3), whose own inputs (the GDP-level gap, the unemployment change) are themselves

4.6 External sector

An external block adds Brazil's exposure to global commodity prices and the exchange rate as a shock common to every scenario within a given Monte Carlo draw, so that differences between scenarios remain attributable to policy rather than to shared external luck — a design chosen deliberately after noting that the 2003–2010 commodity boom and the 2014–2016 recession dominate Brazil's growth variance far more than any plausible fiscal-policy counterfactual. A first specification applied the elasticity of GDP to the terms of trade estimated by Gruss (2014) for a panel of Latin American and Caribbean economies; because that coefficient implies an implausibly small Brazilian response to the observed 2003–2010 boom, a second, Brazil-specific specification estimates the response of GDP growth, the EMBI spread, the real interest rate, the investment rate and commodity-linked dividend revenue to commodity export prices, US interest rates and world growth directly, on two deliberately non-overlapping estimation windows (2000–2014 and a reverse window, 2010–2024), reporting Newey–West standard errors and an explicit out-of-sample test. The reverse-window specification recovers 59% of the observed 2006–2010 boom out of sample, against 3% for the original terms-of-trade specification, and is used, with the coefficient set drawn per Monte Carlo draw from either estimation window with equal probability, as the Monte Carlo's output channel; exchange-rate pass-through to inflation (10% within a year, 16% cumulative over two, following Banco Central do Brasil estimates) and the debt consequences of that inflation are otherwise unchanged. The 2014–2016 domestic recession remains unexplained by every specification tried, a result we report rather than paper over with an additional free parameter.

4.7 Governance: leakage and major-corruption shocks

The model separates two distinct governance phenomena. Routine leakage treats a share of social-service spending, sampled uniformly from 2.8% to 13%, the full range reported by the audited municipalities in Ferraz (Finan and Moreira) (whose own point estimate, around 8% on average, sits near the centre of that range), as diverted before it becomes services; because the same share is applied to every scenario and both candidate programmes alike, it cancels exactly from every scenario-versus-baseline comparison and is reported instead as a separate "leakage halved" sensitivity on the absolute level of health and mortality outcomes. A second, independent module represents rare major-corruption scandals — calibrated on the 1992 Collor impeachment crisis and the 2014–2016 Lava Jato investigation, the only two post-1990 episodes with a measurable macroeconomic footprint — as a stochastic annual event process (3–8% probability a year) with a triangularly distributed GDP-level loss (1.0%/2.25%/3.6%, mode from secondhand GO Associados and Tendências estimates, high end from a contested Dieese estimate), a two-year build-up and a four-year recovery half-life. The same drawn event sequence and magnitudes are applied identically to every scenario and to both 2026 programmes within a draw, so the module widens uncertainty bands without mechanically favouring any one policy path; it is explicitly not a statement about the corruption propensity of any candidate, a point we return to in Section 9 because the perception indices available (the World Bank's Control of Corruption estimate and the Transparency International Corruption Perceptions Index) cannot, within their own measurement uncertainty, distinguish recent Brazilian governments from

outputs of this feedback block (C5), so the ordinary pipeline cannot supply Gini before it is needed; an off-by-default sensitivity run resolves this with a second, pre-feedback C3 pass used only to seed the channel, rather than folding it into the default ordering. Second, and more fundamentally, the Ostry (Berg and Tsangarides) coefficient is a single cross-country IMF staff estimate, not Brazil-specific; consistent with every other borderline channel in this paper (the nonlinear risk premium, the fiscal-reaction-function check, Appendix B's and Appendix C's diagnostics generally), a mechanism resting on one externally-contested, non-Brazil-specific estimate is reported as an available sensitivity rather than folded into the default, published results.

one another.

5 Scenario design and the 2026 candidate programmes

5.1 The core scenario grid

Against the status-quo (S0) and constant-size (S1) baselines, the model defines paired reduction and expansion scenarios at five, ten, fifteen and twenty years: a moderate reduction (R_MOD, 1/2/3/4 pp of GDP below S0) and an ambitious reduction (R_AMB, 2/4/6/7 pp), together with a precedent-based stress case (R_MAX) calibrated from the tenth percentile of observed middle-income-country consolidation episodes of comparable duration; and mirror-image debt-financed expansions (E_MOD, E_AMB) and tax-financed expansions (E_MOD_TAX, E_AMB_TAX) of the same size. Reductions are measured against S0 rather than S1 because ageing itself raises pension and health spending regardless of any discretionary policy choice, and the question of interest is the size of a deliberate reduction on top of that drift, not a reduction relative to an already-adjusted baseline. A default reduction package allocates 40% to pension and benefit indexation, 25% to subsidies and tax expenditures, 20% to administrative and service efficiency, and 15% to other current spending, with public investment and means-tested transfers protected by default; four alternative compositions — pension-only, efficiency-only, an investment-also-cut stress test, and a compensating-transfers variant — are run as sensitivities, because the model's documentation is explicit that the *composition* of a reduction, not merely its size, is the dominant determinant of its growth and distributional consequences.

5.2 The 2026 run-off programmes as scenarios

The same machinery, without modification to a single macro, micro or social equation, is used to translate the two candidates' filed 2026 electoral programmes into comparable annual fiscal paths over an assumed 2027–2030 term, with the resulting package gap held constant thereafter. Three general extensions to the scenario machinery were required and were checked, by byte-identical regression against every existing published scenario, to leave those scenarios entirely unaffected: a per-package annual gap path (rather than a single total multiplied by fixed shares), a debt adjustment applied outside the primary balance (for privatisation proceeds), and a tax-relief allocation rule in the household microsimulation for revenue reductions, mirroring the existing rule for revenue increases.

The incumbent's programme is mapped onto the constitutional spending-growth rule actually in force (Complementary Law 200/2023): real federal primary spending grows at 70% of real revenue growth, bounded between 0.6% and 2.5% a year, applied to the federal share (48%) of general-government primary spending, with an assumed modest expansion of targeted transfers inside the resulting ceiling and a modest revenue ramp from announced measures. In the taxonomy of Kopits and Symansky (1998), this is a textbook expenditure rule, of the kind Cordes et al. (2015) find has a mixed compliance record internationally and Feld and Kirchgässner (2008) find has held durably in the one well-studied case (the Swiss debt brake) where it was paired with a credible escape-clause design; Brazil's own Independent Fiscal Institution flags the rule's reliance on an uncertain, court-dependent revenue base as its principal design weakness (Instituição Fiscal Independente, 2023). The challenger's programme is mapped onto an explicit debt-stabilising rule: the primary balance moves linearly from its 2026 status-quo level to the level $d\,(i-g)/(1+g)$ that would stabilise — not reduce — the 2026 debt ratio at status-quo interest and growth rates, over an assumed two- to six-year horizon, financed partly by an assumed path of privatisation proceeds retiring up to 2 pp of GDP of debt over the term. This is, in the same taxonomy, a balance rule, the type López-Villavicencio and Zoumenou

(2025) find is the only rule type robustly associated with improved fiscal outcomes in a recent European Union panel, for whatever that external validity is worth in a Brazilian context; both rules sit alongside, rather than replace, the pre-existing constitutional "golden rule" restricting borrowing to capital expenditure, whose own effectiveness Pires (2019) and Nunes et al. (2020) separately question. For both programmes, the pooled historical median bias between announced fiscal targets and realised outturns, +0.25 pp of GDP (outturn looser than target; Section 5.3), is added to the planned gap from 2027 onwards, and every programme-specific magnitude left unstated by the candidates' own documents is sampled uniformly over an explicitly documented range on a dedicated random stream, so that this additional source of uncertainty does not contaminate the parameter draws used for every other scenario.

We emphasise two points about this exercise that bear on how its results should be read, and why we do not report headline programme-comparison figures in this paper. First, both mappings are judgment calls about how to translate qualitative or partially specified campaign commitments into a fiscal path; neither candidate's programme specifies, for instance, the exact composition of spending cuts implied by its stated ceiling, so the model's own default composition assumption necessarily does much of the work in determining the output and distributional consequences, and a common-composition sensitivity run is required alongside the main run to separate the effect of size and timing from the effect of an assumed composition. Second, and more fundamentally, the model's own historical back-test (Section 7) fails its stated tolerance for the absolute level of the debt-to-GDP ratio, and our internal publication protocol withholds any debt-level result, programme-specific or otherwise, until that gate is cleared or the result is explicitly and prominently qualified as failing it. We therefore present the programme-mapping methodology here as a worked demonstration of the scenario machinery's generality — any sufficiently specified fiscal rule, legislated or proposed, can be run through the same chain — without presenting unpublished headline comparisons as findings.

5.3 Fiscal delivery: targets against outturns

Because both candidate mappings rely on an assumed fiscal target being approximately met, we separately assembled a nineteen-year panel (1999–2025, with several years unavailable in the archived annexes, leaving thirteen usable consolidated-public-sector observations) of original and revised primary-balance targets from the Budget Guidelines Law (LDO) annexes against realised outturns, cross-checked against the Central Bank's public-sector balance series. Pooled across governments and excluding the pandemic year's suspended target, the median miss against the original target is +0.25 pp of GDP (outturn looser than target; mean +0.09 pp, p10 −2.69, p90 +2.81), with wide dispersion by period and a mean sensitive to a small number of large deviations (notably the 2021–2022 post-pandemic years). No government period is treated as better or worse than another in this exercise: differences between periods are generally within the range one would expect from the small number of observations per period, and the document explicitly refrains from ranking governments on this basis.

6 Monte Carlo propagation of uncertainty

For each of one thousand draws (seed 20261020, fixed for reproducibility), every ranged parameter in the model — fiscal multipliers, the real interest rate, total-factor-productivity and human-capital growth, the investment rate, the Okun coefficient, the pension indexation weight, the health-spending service-cut fraction, the infant-mortality elasticity, the revenue elasticity, the health-to-output and public-capital elasticities, the sovereign-risk-premium coefficient, the social-spending leakage share, and the household-microsimulation pension, labour and tax-incidence variants — is sampled once from its documented range (triangular where a central

estimate is available, uniform otherwise) and the entire C1–C2–C4 chain is then re-run deterministically at that single sampled value, so that no model equation is duplicated or approximated for the stochastic run: the Monte Carlo differs from the published deterministic corners only in which point of each parameter's range it evaluates. Differences against the status-quo scenario are computed *within* each draw before taking percentiles across draws, so that a reported interval describes the uncertainty in the *policy effect* under shared parameter uncertainty, not the (much wider) gap between two independently resampled clouds. Five package compositions and two multiplier-persistence settings are run for every scenario. The household microsimulation, long-run feedback channels, external shocks and major-corruption shocks are added as further, separately labelled columns rather than folded silently into the core percentiles, because each carries materially different and sometimes much weaker evidentiary support than the core macro-fiscal chain; a user who wants only the best-supported result can read the core columns and ignore the rest.

We are explicit, following the model's own documentation, about what these percentiles do not represent. They describe uncertainty in the model's *parameters only*: they are not forecast intervals, they do not include model-specification uncertainty, they assume no correlation between sampled parameters because no credible joint distribution was available, and the triangular and uniform sampling shapes are themselves a modelling choice rather than an estimated distribution. External shocks and the major-corruption process are included only in columns explicitly labelled as such, precisely so that a reader cannot mistake the width of the core policy-uncertainty bands for the width of total outcome uncertainty.

7 Historical back-testing and the limits of validation

Before any scenario result is treated as fit for use, the model is checked against Brazil's own recent fiscal and macroeconomic history, using a two-tier gate agreed in advance of seeing the results. **Tier 1** asks whether the model's equations reproduce observed year-on-year *changes* with acceptable error: the debt-accumulation identity, evaluated one step at a time from each year's observed starting debt, has a mean absolute error of 2.05 pp of GDP over 2011–2024 (tolerance 2.5 pp); the growth-accounting block has an absolute mean error of 0.165 pp a year over 2006–2019 (tolerance 0.5 pp); and the external output channel's reverse-window specification achieves an out-of-sample root-mean-squared error of 3.00 pp against a naive-forecast benchmark of 3.88 pp over 2015–2024. All three pass. **Tier 2** asks a harder question: does a debt projection, anchored once in 2017 and then run forward seven years using only the model's own equations, reproduce the observed 2024 debt level? It does not: the recursion overshoots observed 2024 debt by 16.1 pp of GDP with no correction, and still by 12.5 pp of GDP even after applying the best available historical correction (the 2011–2017 mean implied stock-flow adjustment). Figure 2 shows this failure directly: the residual against both reconstructions widens in every single year from the 2017 anchor, not only after some later break point, and the divergence is visually starkest after 2020 — observed debt peaks with the pandemic and then falls and stabilises, while both reconstructions keep climbing every year through 2024, since the model has no channel that would turn a recursion downward once primary balances and growth are fed through it mechanically. Tier 2 fails, and the model's own publication protocol accordingly withholds any absolute debt-level result from being presented as validated, restricting published results to paired scenario *differences* against the status-quo baseline, which is precisely what Tier 1 was designed to certify.

We consider this an important result in its own right, not merely a caveat. The residual is not evenly distributed: the single largest contributor is a set of discretionary Treasury-to-BNDES lending and BNDES early-repayment and Central Bank result-transfer operations

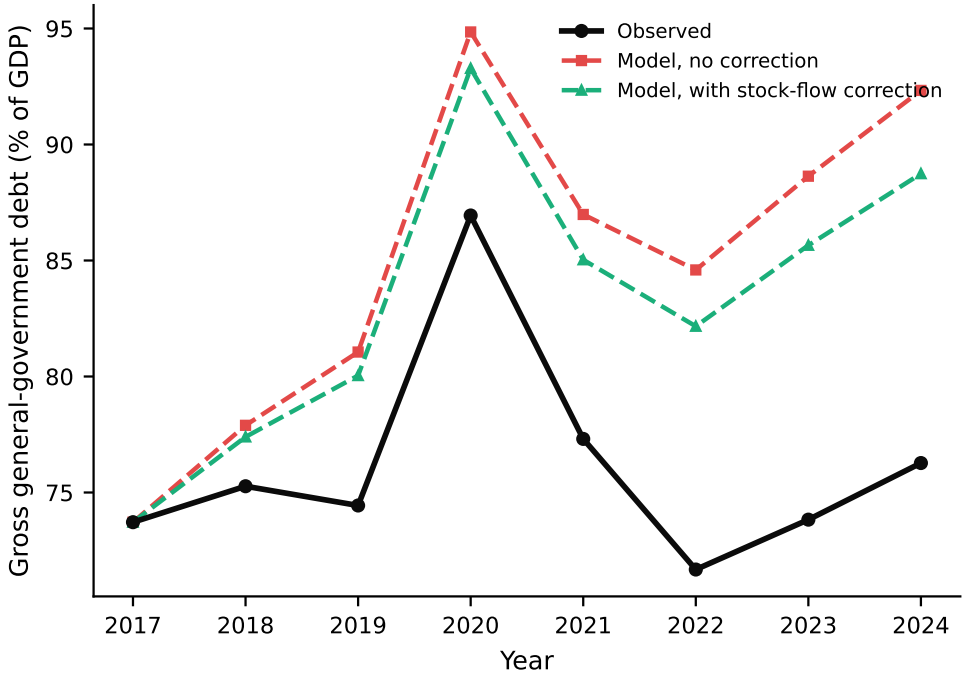


Figure 2: Tier 2: observed Brazilian gross general-government debt/GDP, 2017–2024, against the model's own debt-identity recursion anchored once in 2017, with and without the 2011–2017 mean stock-flow-adjustment correction. Neither reconstruction recovers the observed post-2021 decline; both overshoot by 2024 (16.1 pp and 12.5 pp of GDP respectively).

between 2011 and 2022, which move the implied stock-flow adjustment by as much as 3.9 pp of GDP in a single year and show no relationship to any macroeconomic covariate — growth, interest rates, the terms of trade or the sovereign spread — available to the model. These are discretionary policy decisions, not quantities any forward-looking macro model, however well specified, could have predicted several years in advance, a description that fits squarely within the broader literature on quasi-fiscal operations of public financial institutions (Mackenzie and Stella, 1996) and on stock-flow adjustments as a large, persistent and only partly explicable source of debt accumulation worldwide (Weber, 2012; Seiferling, 2013): Jaramillo (Mulas-Granados and Kimani)'s finding, from 179 historical debt-spike episodes, that spikes driven by stock-flow adjustments rather than primary deficits predict debt subsequently *failing* to decline, is a genuinely sobering precedent for how Brazil's own episode should be read (the fuller positioning of this literature, including the closest peer-reviewed treatment of the Brazilian episode specifically, is in Sections A.10 and A.11). Rather than search for a better-fitting correction, which would mean fitting noise, we instead represent this uncertainty honestly: a stochastic stock-flow-adjustment term, bootstrapped in three-year blocks from the demeaned 2011–2024 historical series and applied identically to every scenario within a Monte Carlo draw (so that it widens absolute debt-level uncertainty without mechanically favouring any policy), is added to the published Monte Carlo as of the most recent run. This does not and cannot make the point-forecast back-test pass; it instead ensures that published debt-level *ranges*, where they are shown at all, are not falsely narrow. We regard the choice between a better point estimate and an honestly wider interval, when the former is simply unavailable, as itself a methodological contribution worth stating plainly: a model that cannot forecast a quantity should say so by widening its uncertainty, not by implicitly asserting precision it does not have.

A further internal review identified that the growth-block Tier 1 check had originally reused a fixed total-factor-productivity growth rate estimated over 2000–2019, a window that fully contains the 2006–2019 test period, so that its pass demonstrated internal consistency rather than held-out predictive skill, unlike the debt-identity check (a pure accounting relationship with no fitted parameter) and the external-channel check (fitted on 2000–2014, tested on 2015–2024 only). This has since been corrected: the growth-block statistic is now estimated on a genuinely pre-sample total-factor-productivity window (1996–2005, ending seven years before the test period begins) and the Tier 1 gate is re-keyed to this corrected statistic, which the model passes more comfortably than the original, contaminated one (mean absolute error −0.036 against 0.165 pp of GDP a year). We report the original defect, and its fix, explicitly, because it is precisely the kind of in-sample contamination that a less self-critical validation exercise would allow to pass unremarked, and because a corrected statistic that happens to pass more easily than the flawed one it replaces is itself worth distinguishing from a corrected statistic that was simply re-engineered until it passed.

7.1 Diagnostic extensions: a brief summary

Tier 2's failure leaves open a question the back-test itself cannot answer: how much of the headline status-quo debt path is an artefact of the model's deliberately simple, no-policy-reaction specification, and how much would a more behaviourally realistic government, or a more aggressively nonlinear market response, change it? Appendix B reports, in full, a set of clearly labelled, off-by-default diagnostics built to answer exactly this, run only against the status-quo baseline (S0) and never folded into any published Monte Carlo summary. In brief: under the model's own preferred specification, including the stochastic stock-flow-adjustment term of Section 7, the probability that S0's debt ratio exceeds 90% of GDP is already 73.6% at five years and 95.0% by ten; a fiscal-reaction-function variant using Brazil's own, counter-intuitively signed estimated coefficient produces a *higher*, not lower, debt path than the unconditional baseline; and an explicitly nonlinear, fiscal-fatigue-style risk-premium variant, following Ghosh et al. (2013) and Bi (2012), stays close to the model's existing linear premium below a historical threshold but diverges sharply above it. Every absolute figure in Appendix B inherits the same Tier 2 bias documented above and should be read accordingly, as an upper-bound diagnostic rather than a calibrated forecast.

A second, independent review raised a different question entirely: not whether the status-quo debt path is realistic, but whether the paired scenario *differences* of Table 1 themselves are biased by channels the model omits (the business cycle, a monetary reaction function, the exchange rate and the distribution of interest income). Appendix C reports six further off-by-default diagnostics built in direct response, and Section 9 gives the resulting answer in full.

8 Headline results

We report here only indicators that have cleared the Tier 1 gate, as paired differences against the status-quo baseline; absolute debt levels are reported as differences versus S0, never as validated levels, consistent with Section 7. Readers wanting the fuller, Tier-2-honest picture of S0's own absolute debt level — breach probabilities, the fiscal gap, and the fiscal-reaction-function and nonlinear-premium diagnostics — should turn to Appendix B, which is deliberately not folded into the paired-difference table below. All figures are the median across one thousand draws, with the 10th–90th percentile range in brackets, for the default package composition and the transitory-multiplier persistence setting unless stated otherwise.

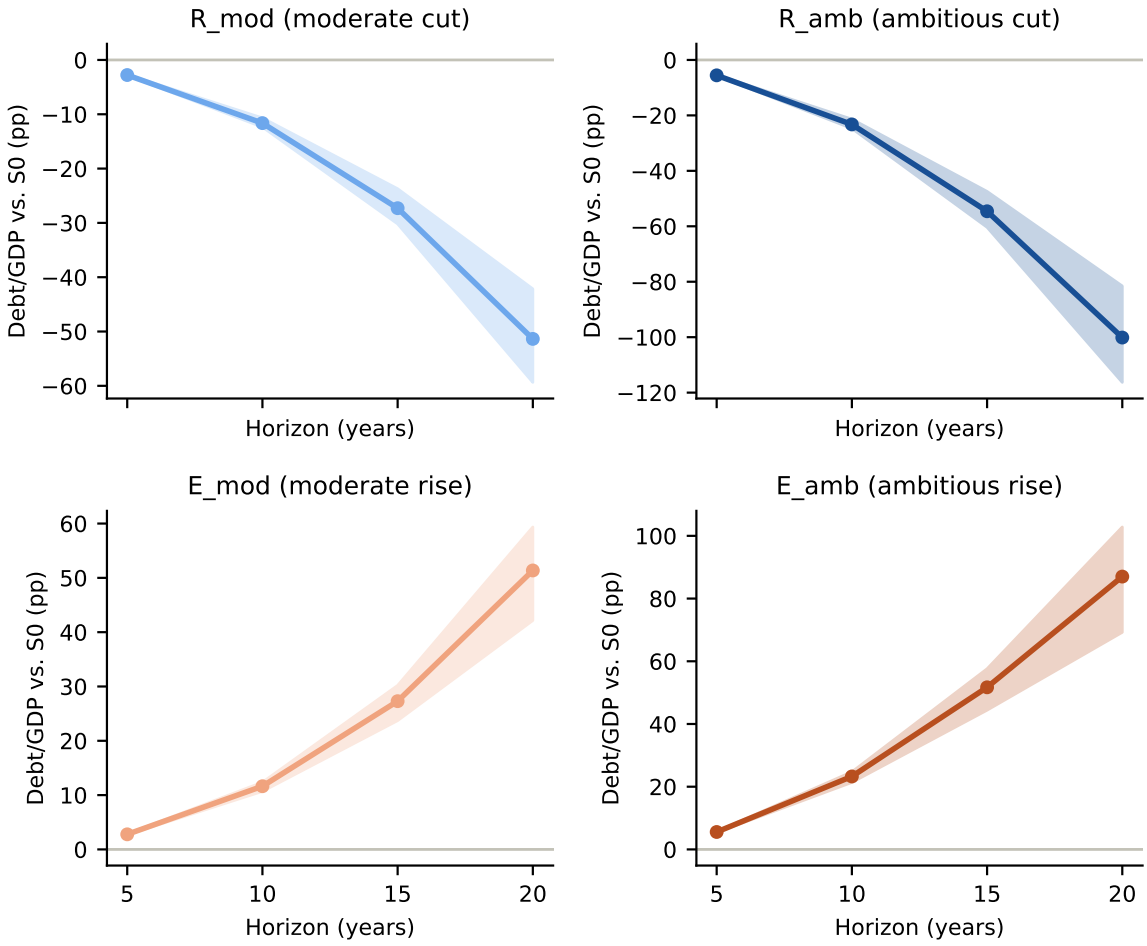


Figure 3: Debt/GDP vs. S0, by scenario and horizon: median (line) and parameter-uncertainty band (shaded, 10th–90th percentile) across one thousand draws, default composition, transitory multiplier persistence. Reported only at the model's four named horizons (5/10/15/20 years), joined for legibility, not a continuous forecast path — the bands describe resampled parameter uncertainty, not a conditional quantile forecast (Section 6).

Table 1: Selected ten-year (2035) scenario effects against the status-quo baseline, default composition, transitory multiplier persistence. Median [p10, p90] across 1,000 Monte Carlo draws.

Indicator	R_MOD (moderate cut)	R_AMB (ambitious cut)	E_MOD (moderate rise)	E_AMB (ambitious rise)
GDP level vs. S0 (%)	−0.40 [−0.48, −0.31]	−0.79 [−0.97, −0.62]	+0.40 [+0.31, +0.48]	+0.79 [+0.62, +0.97]
Debt/GDP vs. S0 (pp)	−11.6 [−12.4, −10.7]	−23.3 [−24.7, −21.4]	+11.6 [+10.7, +12.4]	+23.3 [+21.4, +24.7]
Unemployment vs. S0 (pp)	+0.07 [+0.06, +0.09]	+0.14 [+0.11, +0.17]	−0.07 [−0.09, −0.06]	−0.14 [−0.17, −0.11]
Infant mortality vs. S0 (per 1,000)	+0.035 [+0.017, +0.055]	+0.069 [+0.034, +0.110]	−0.035 [−0.056, −0.017]	−0.070 [−0.112, −0.035]
Gini vs. S0 (points)	+0.16 [+0.13, +0.57]	+0.36 [+0.33, +1.27]	−0.10 [−0.46, −0.08]	−0.16 [−0.83, −0.11]

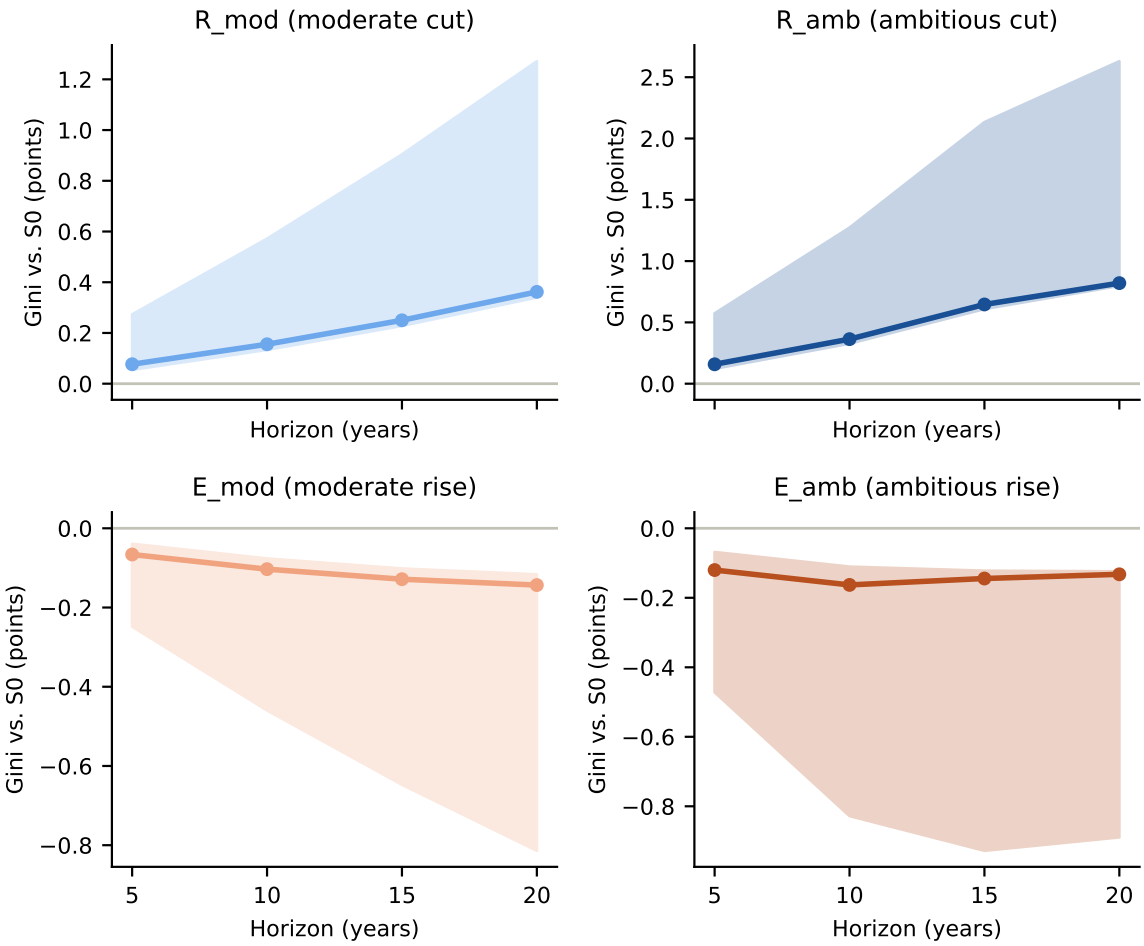


Figure 4: Gini coefficient vs. S0, by scenario and horizon, same data and convention as Figure 3.

Three features of these results are worth drawing out, visible directly in Figures 3 and 4 as well as in Table 1. First, the central ten-year reduction in debt from an ambitious, sustained 4 pp-of-GDP consolidation is a little under 24 pp of GDP against the status-quo baseline, roughly in line with the arithmetic of the debt identity rather than requiring any output-side heroics; Figure 3 shows every scenario's p10–p90 band widening by a strikingly consistent factor of roughly ten between the ten- and twenty-year horizons (9.9–10.3× across all four scenarios), so parameter-uncertainty growth here tracks horizon rather than policy ambition. The GDP cost of achieving the consolidation is under one percentage point at the median under transitory multipliers, but rises to a median GDP loss of 3.9% at ten years under the persistent-multiplier assumption (Section 4), a difference large enough that no single headline number for "the cost of consolidation" is defensible without stating which multiplier persistence assumption it uses. Second, the status-quo debt path itself, with no policy change, rises from 78.6% to a central 145.5% of GDP by 2035, of which 61% of the ten-year increase is attributable to the interest-growth differential rather than to the (ageing-driven) widening of the primary deficit; this is the arithmetic fact that gives any debt-stabilisation rule, of the kind proposed in the challenger's 2026 programme, its force, independently of whether the specific numerical target chosen is the right one. Third, every reduction scenario raises the simulated Gini coefficient and infant mortality, and every expansion scenario lowers both, by magnitudes consistent in direction, if not in exact size, with the cross-country consolidation literature (Ball et al., 2013) and the Brazilian health-spending elasticity literature (Szklo, Clarke and Rocha) on which the two blocks separately draw — a consistency we read as mildly reassuring given that neither block was calibrated to reproduce the other's finding. Figure 4 shows this holds up under the model's own parameter uncertainty, not only at the median: none of the four scenarios' p10–p90 Gini bands cross zero at any reported horizon, so the sign of the headline Gini effect is robust to the sampled parameters this Monte Carlo varies. The reversal discussed in Section 9 comes from a channel this Monte Carlo does not sample at all — interest income's incidence, added only in Appendix C — not from parameter uncertainty within the published run.

For a reader who wants the policy-relevant content of this paper without the two 2026 programmes' withheld headline comparison, these three findings are it: the composition and persistence of any fiscal adjustment matter at least as much as its headline size; the interest-growth differential, not discretionary policy, is the arithmetic force driving status-quo debt upward, which is what gives a debt-stabilisation rule of the kind either candidate might adopt its logic independently of the specific target chosen; and any reduction large enough to bend the debt path materially also moves distributional and health outcomes in the same direction, by an amount this paper reports rather than assumes away.

9 Discussion: what this model is, and is not, for

Three limitations deserve emphasis beyond those already noted in the discussion of each block.

This is a comparative-statics and accounting instrument, not a dynamic stochastic general-equilibrium model, and the omitted channels plausibly, though not demonstrably, bias the expansion scenarios' reported distributional improvement. Prices, wages and behaviour do not adjust endogenously to a change in fiscal policy except through the specific, named channels described above (a fiscal multiplier, an Okun-law unemployment response, a household's mechanical exposure to a given tax or transfer change). There is no labour-supply response, no portfolio reallocation, no monetary-policy reaction function, and no general-equilibrium price adjustment. This is a deliberate choice: each of those omitted channels is itself contested in the literature, and we judged it more honest to omit a channel entirely, and say so, than to include a stylised version calibrated to no particular Brazilian estimate. But the omission is

not symmetric in its likely consequence, and an independent review of this paper made exactly this point: a sustained, debt-financed expansion would plausibly trigger a connected chain this model does not price at any link — a higher policy rate as the Central Bank responds to the resulting inflation and debt trajectory, a weaker real as fiscal stress widens the sovereign spread, higher borrowing costs crowding out private investment, an inflation tax that falls hardest on poor households' largely non-indexed cash holdings, and higher interest payments accruing disproportionately to the bondholders who are concentrated in the top income centiles. Every one of these channels runs in the direction of raising, not lowering, the measured Gini and poverty cost of an expansion, which is why the model's own finding that expansions mechanically lower the Gini coefficient (Section 8) should be read as the effect of the transfer and tax channels *alone*, holding monetary policy, the currency and private portfolio income fixed, not as a net welfare statement. Unlike the general-equilibrium channels we have not attempted to quantify, two of these specific channels — interest income distributed the way it is actually concentrated, and a Taylor-rule Selic response to a scenario's own output gap — are quantified in full as off-by-default diagnostics in Appendix C, run separately from, and never folded into, Table 1. The answer is not a hedge: combining each diagnostic's own median effect with the published figure (Section C.6) moves E_MOD's and E_AMB's median ten-year Gini effect from −0.10 and −0.16 points to +0.65 and +1.15 points respectively — the distributional finding *reverses sign* once interest income is distributed correctly, and the debt cost of both scenarios is understated by roughly 13–14%. We regard this as settling the direction of the reviewer's claim, for the specific channels we could quantify, while still not settling the channels we could not (the labour-supply and portfolio-reallocation responses named above remain genuinely unquantified).

Public debt is modelled as a single undifferentiated stock, with no split by indexation, currency or holder, though the composition this omits is, for Brazil specifically, concentrated in a domestic-rate risk the model already prices rather than a currency risk it does not. The debt-accumulation identity (Section 4) applies one interest rate to the entire debt stock; it does not distinguish Selic-linked floating-rate bonds from inflation-linked, fixed-rate or foreign-currency-linked instruments, and the exchange rate (Section 4) affects debt only through inflation pass-through, never through direct revaluation of foreign-currency liabilities. This is a genuine simplification, but its practical consequence for Brazil is smaller than for a dollar-indebted emerging market: on the Brazilian National Treasury's own most recent published composition, Selic-linked (floating-rate) instruments are both the largest share of federal domestic debt and a rapidly rising one (48.3% at the close of 2025, 52.7% by August 2026, against a revised 2026 target band of 49–53%), inflation-linked instruments a further fifth to a quarter, fixed-rate instruments a fifth, and foreign-exchange-linked instruments a stable 3.7–4.4% of the total, a range the Treasury's own borrowing plan keeps essentially unchanged year on year (Secretaria do Tesouro Nacional, RMD). Separately, roughly 96% of federal debt is placed in the domestic market (Secretaria do Tesouro Nacional, RMD). The composition risk this model therefore omits is overwhelmingly the one it already prices through the debt identity's single interest rate and the sovereign-risk-premium channel, namely a rise in the Selic (the Treasury's own published sensitivity: a 1 pp rise in the Selic adds approximately R$60.6 billion, about 0.46% of GDP, to gross debt in year one alone, Secretaria do Tesouro Nacional, RMD), not the foreign-currency revaluation risk a multi-currency debt model would be built to capture. We quantify this valuation adjustment directly in Appendix C (Tables 5 and 6): it is indeed small at this paper's main ten-year horizon, confirming the framing above, but grows to a median 5–17 pp of GDP by twenty years under the common bootstrapped exchange-rate path, and remains small (under 0.5 pp at twenty years) even once the depreciation itself is made to respond to a scenario's own fiscal stress rather than drawn unconditionally — confirming that this channel stays small because the underlying FX-linked share is small, not because the

mechanism linking fiscal stress to the exchange rate is inert.

The model cannot and does not attempt to assess the corruption propensity, competence, or governing quality of any candidate or government. Both the routine-leakage parameter and the major-corruption shock process are, by explicit design, applied identically to every scenario and to both 2026 programmes within a Monte Carlo draw; this is not a simplifying assumption adopted for convenience, but a direct consequence of the fact that the only available perception-based measures of Brazilian governance quality — the World Bank's Control of Corruption estimate and the Transparency International Corruption Perceptions Index — cannot, within their own stated measurement uncertainty, distinguish recent Brazilian administrations from one another. A model that tried to assign different corruption parameters to different candidates on the basis of these indices would be manufacturing precision the underlying data does not support.

The model's single most consequential finding may be about its own limits rather than about fiscal policy. The Tier 2 back-test failure documented in Section 7 is a substantive result: a model built from a textbook debt-accumulation identity, careful growth accounting, and an explicitly estimated external-sector channel still cannot reproduce six years of observed Brazilian debt accumulation to within five percentage points of GDP, because a material part of that accumulation is driven by discretionary balance-sheet operations between the Treasury, the development bank BNDES and the Central Bank that have no macroeconomic antecedent a model of this kind could observe in advance. This is not a peculiarly Brazilian failing: the wider stock-flow-adjustment literature surveyed in Section A.10 finds the same pattern — large, persistent, poorly-explained residuals — across more than a hundred countries (Weber, 2012), and finds that large residuals of exactly this kind predict debt subsequently failing to decline (Jaramillo, Mulas-Granados and Kimani), which is precisely why we chose to widen the model's uncertainty bands rather than search for a better-fitting correction. Any debt-sustainability exercise for Brazil — ours or another's — that reports a single confident debt-level projection without confronting this fact should, we think, be read with considerable scepticism.

The Monte Carlo reports a resampled distribution, not a true conditional quantile forecast, and the distributional block omits a fiscal-impoverishment measure. The percentiles reported throughout this paper, including the Appendix B breach probabilities, describe the spread induced by resampling independently drawn structural parameters; they are not the output of a quantile regression conditioned on the current state of the economy, in the sense the growth-at-risk and debt-at-risk literature now uses the term (Prasad et al., 2019; Lian, Presbitero and Wiriadinata; Furceri et al., 2025). Extending the model to condition its percentiles on observable pre-shock conditions, rather than treating every draw as exchangeable, is a natural next step we have not undertaken. Separately, the household microsimulation reports the Gini coefficient and two poverty headcounts but not the fiscal-impoverishment measure of Higgins and Lustig (2016), which would show directly whether any household is left poorer in absolute terms by a package that is, on the Gini and headcount measures reported here, both progressive and poverty-reducing; closing this gap would strengthen the distributional claims made in Section 8 without requiring any new data source.

10 Conclusion

We have described a modular, stochastic framework for evaluating changes in the size and composition of the Brazilian state, chaining a demographic fiscal baseline, a growth-and-debt macro block, a household microsimulation, a social-outcomes block, a set of long-run feedback channels, an external-sector block, and an explicit governance layer, propagated through a thousand-draw Monte Carlo and checked, as far as the available data permit, against Brazil's

own fiscal history. Applied to a common grid of reduction and expansion scenarios, the model finds that the composition and persistence assumptions attached to a given fiscal adjustment matter at least as much as its headline size; that population ageing alone will raise age-related spending by several percentage points of GDP over the coming decade regardless of policy; that the interest-growth differential, not the primary balance, is projected to be the dominant driver of status-quo debt accumulation; and that the model's own historical back-test, read honestly, fails to validate any single confident debt-level projection, a failure we trace to discretionary balance-sheet operations no macro-fiscal model of this design could have anticipated, and which an independently documented, cross-country stock-flow-adjustment literature suggests is a structural feature of sovereign debt accounting rather than a peculiarly Brazilian one. In direct response to that failure, we report — as a clearly labelled, off-by-default diagnostic never folded into the headline results — how much the status-quo debt path would change under a typical emerging-market fiscal-reaction coefficient, under Brazil's own, oppositely-signed estimated coefficient, and under an explicitly nonlinear, fiscal-fatigue-style risk premium, together with EC/IFI-style breach probabilities and a fiscal-gap statement that a bare median debt path cannot provide. Applied, as a methodological demonstration, to the two candidate programmes filed for Brazil's 2026 presidential run-off, the same machinery shows that a constitutional spending-growth rule and an explicit debt-stabilising rule are, formally, two different closure rules for the same debt-accumulation identity, whose consequences depend on assumptions — about composition, about delivery against stated targets, about privatisation proceeds — that neither candidate's own published programme fully specifies; we withhold the resulting headline comparison from this paper pending the same internal validation gate applied to every other result. We offer the framework itself, and the discipline of its parameter registry and its publication gate, as the paper's principal contribution: a fiscal scenario model is, in our view, more useful for the questions it refuses to answer with false precision than for the ones it answers confidently.

Read only as an accounting exercise, not a recommendation, four things the model's own numbers imply are worth weighing in the next term. The interest-growth differential, not spending size, drives most of the status-quo debt increase (Section 8), so credibility that compresses the sovereign-risk premium would plausibly do more than any single consolidation's headline size (Blanchard, 2019; Furman and Summers, 2020). The model's own package-composition sensitivities, consistent with the wider composition-of-consolidation literature (Alesina and Ardagna, 2010; Pires, 2014), suggest protecting public investment and administrative efficiency ahead of pension indexation would moderate any adjustment's growth and distributional cost. Ring-fencing health spending has a quantified, not merely plausible, payoff for child mortality (Szklo, Clarke and Rocha; Rasella et al., 2013). And a debt-financed expansion's distributional benefit is not automatic: Appendix C shows it depends on whether the resulting interest income is left with its actual, concentrated holders (Lustig, 2018; Higgins and Pereira, 2014) or paired with an instrument that claws some of it back. None of this ranks the two 2026 programmes (Section 5) or forecasts what either would do; it is what this paper's own quantified trade-offs imply, stated with the same hedges attached to every other number in this paper.

Data and code availability

The code, configuration, and the complete parameter registry required to reproduce every result in this paper, together with a README describing how each table and figure is generated, are openly available in the public GitHub repository `https://github.com/AugustoFM/minimal-state-world-dashboard`, hosted under the author's own personal account (the handle "AugustoFM" combines his middle and first names with his surname initial, in that order). A

permanent tag or release pinned to the exact commit underlying this submitted version will be created at the time of submission and referenced here, so that reproduction of the results in Tables 1, 2, 3 and 4 does not depend on the repository's actively developed main branch. The repository includes: the full simulation codebase (data preparation, macro, microsimulation, social, feedback, external and Monte Carlo modules); the single machine-readable parameter registry recording, for every numerical input used in any published result, its source, vintage, licence and a verification flag; the task-by-task development log (`tasks/STATUS.md`) documenting each modelling decision and its review; and the scripts that generate Tables 1, 2, 3 and 4 directly from the committed Monte Carlo output. Raw PNAD Contínua microdata are not redistributed in the repository, in line with IBGE's own distribution terms, but are freely downloadable from IBGE and the repository's data-preparation scripts reconstruct the exact extract used here from that public source. No other dataset used in this paper is proprietary or restricted.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. The author is an independent researcher and undertook this work without institutional affiliation or external funding.

Conflict of interest

The author declares no conflict of interest. The author has no financial, professional or personal relationship with either candidate or campaign referenced in Section 5, and the programme-mapping exercise described there is explicitly withheld from headline publication pending the model's own internal validation gate, a decision made for methodological rather than political reasons.

Ethics statement

This study uses only secondary, publicly available official statistics and anonymised government survey microdata (IBGE's PNAD Contínua); it involves no original data collection from human participants, no identifiable personal data, and no intervention. No ethics committee approval was required or sought.

Author contributions

Flaviano A. Moreira: Conceptualisation; Methodology; Software; Validation; Formal analysis; Investigation; Data curation; Writing – original draft; Writing – review and editing; Visualisation; Project administration. (CRediT taxonomy; sole author.)

Acknowledgements and disclosure of AI-assisted tools

In accordance with the publisher's policy on generative artificial intelligence, the author discloses the following uses of AI technologies in producing this paper and the research underlying it. *Research methodology, software development and data analysis:* the simulation codebase (approximately fifteen thousand lines across the data-preparation, macro, microsimulation, social, feedback, external and Monte Carlo modules) was developed iteratively with the assistance of AI coding agents, principally Anthropic's Claude Code (Claude Opus and Sonnet model

families, 2026) and OpenAI's Codex (GPT-5-series models, 2026), operating throughout under the author's direct and continuous direction. The author specified every modelling choice and equation, reviewed and tested all AI-generated code, and is solely responsible for the model's design, its validation (Section 7), and its limitations. *Literature review:* the extended literature review in Appendix A was compiled with the assistance of AI research agents that conducted targeted web searches to locate and verify candidate citations; every citation retained in this paper was independently checked by the author against a publicly accessible primary or indexing source (the publisher's own page, NBER, IMF, RePEc/IDEAS, or an equivalent repository) before inclusion, and candidate citations that could not be so verified were excluded rather than retained on the strength of the AI agent's own assertion. *Manuscript drafting and editing:* AI language-model assistance (the same Claude model family) was used in drafting and iteratively revising the prose of this manuscript under the author's direction; the author reviewed, edited and takes full responsibility for the final text, its arguments, and its conclusions. No AI tool is credited as an author, consistent with the principle that authorship requires the capacity to take responsibility for the work, which these tools do not have. Every numerical parameter used in any published result is independently recorded, with its source, vintage, licence and a verification flag, in the machine-readable registry described in the Data and Code Availability statement above; parameters without a traceable source are marked explicitly as such and excluded from headline results. The author thanks no individual funder or institution, consistent with the Funding statement above.

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A Extended literature review

This appendix gives the full discussion summarised in Section 2, moving from the broad macro-fiscal debates that motivate the model's overall design to the narrow, often Brazil-specific estimates that calibrate individual blocks. Four of the citations below (Levy Yeyati and Sturzenegger, 2023; Lago-Peñas, Cadaval-Sampedro and Herrero-Alcalde; Hanousek and Kočenda, 2011; Balladares and García-Miralles, 2026) were themselves published in *Fiscal Studies*, one of several applied public-finance journals whose recent work bears directly on this paper's own sub-literatures.

A.1 The composition of fiscal consolidation

A long empirical literature asks whether fiscal consolidations differ in their output cost according to whether they are achieved through spending restraint or revenue increases. Alesina and Ardagna (2010) and the subsequent plan-based approach of Alesina et al. (2015,?) find that multi-year adjustment plans built mainly from spending cuts produce markedly milder and shorter output losses than plans built mainly from tax increases, a gap concentrated in private investment rather than consumption, and robust to whether the economy enters the adjustment in recession (Alesina et al., 2017). This finding is contested: Guajardo (Leigh and Pescatori) use Blanchard–Perotti-style narrative identification of exogenous consolidation shocks to reject the Alesina–Ardagna "expansionary austerity" hypothesis more broadly, finding both spending- and tax-based consolidations contractionary once endogenous and anticipated fiscal actions are stripped out, with any residual composition asymmetry a secondary rather than central finding of their design; Perotti (2013) and Jordà (Schularick and Taylor) likewise caution that composition effects in the Alesina–Favero–Giavazzi tradition may partly reflect omitted monetary accommodation and the specific European sample. The present model does not take a position in this debate: it inherits whatever composition the user specifies, pension indexation, subsidies, administrative efficiency, public investment, or means-tested transfers, and reports the growth, debt and distributional consequences of that composition using Brazil-specific multipliers, rather than importing a foreign average effect.

A.2 Fiscal multipliers: size, state-dependence and emerging-market evidence

A logically prior question to the composition debate is how large a fiscal multiplier is in the first place, and whether it varies with the state of the economy. Auerbach and Gorodnichenko (2012) develop the regime-switching methodology that established multipliers are substantially larger in recessions than expansions in advanced economies. Ramey and Zubairy (2018) partially qualify that result on a longer US historical sample, finding multipliers below unity regardless of the degree of slack once defence-spending identification and a longer sample are used, while Ramey (2011, 2019) survey the resulting range, broadly 0.8 to 1.5, and the post-2008 renaissance in multiplier estimation more generally. The closest cross-country precedent to this model's own package-specific approach is Ilzetzki (Mendoza and Végh), who show multipliers vary systematically with exchange-rate regime, trade openness, public debt and the level of development, with emerging, relatively closed, high-debt economies such as Brazil tending towards the lower end of the range. European growth-forecast errors during 2010–2011 austerity, Blanchard and Leigh (2013) show, are consistent with multipliers considerably larger than the then-standard 0.5 assumption, a finding with obvious relevance to the persistent-versus-transitory multiplier

distinction drawn throughout this paper; Nakamura and Steinsson (2014) supply the leading regional-multiplier identification strategy as a cross-check on these aggregate time-series estimates. Two IMF working papers estimate multipliers specifically for Latin American and Caribbean economies: Carrière-Swallow (David and Leigh) and Restrepo (2020) both find multipliers below one, with Restrepo's country-by-country estimates also revealing substantial heterogeneity even within the region, a caution against assuming Brazil's multiplier resembles its neighbours' any more closely than it resembles an advanced-economy average. Born (Müller and Pfeifer) connect multiplier state-dependence to the sovereign-risk-premium channel used in Section 4.5, showing that whether austerity narrows or widens spreads depends on the fiscal starting position. Two Brazil-specific estimates beyond Sanches (2026) and Carvalho and Sanches (2019) corroborate the order of magnitude used here: Pires (2014) estimates a Markov-switching, regime-dependent multiplier for Brazil using IPEA-affiliated methods, and Resende and Pires (2021) find a low aggregate multiplier but markedly higher public-investment and targeted-transfer multipliers, consistent with this model's category-specific rather than single-number approach.

A.3 Stochastic and probabilistic debt-sustainability analysis

The deterministic debt-sustainability exercises conducted by the IMF and finance ministries have long been criticised for reporting a single central path with no account of the uncertainty around growth, interest rates and the primary balance (IMF, 2003). Celasun (Debrun and Ostry) proposed replacing the deterministic path with a stochastic simulation, in which a vector autoregression on the macro-fiscal block generates a joint distribution of shocks that is then propagated through the debt-accumulation identity to produce fan charts of future debt ratios. An important direct antecedent applies closely related risk-management machinery to Brazilian sovereign debt specifically: Garcia and Rigobon (2004) show that a debt path can look sustainable at its central projection while still carrying a non-trivial probability of an unsustainable path once the full joint distribution of shocks is considered, precisely the distinction this paper's own Tier 1/Tier 2 gate (Section 7) is built to respect. Karam and Hostland (2005, 2006) and Mendoza and Oviedo (2004), the latter applying a probabilistic fiscal-solvency framework to Brazil, Colombia, Costa Rica and Mexico directly, are the other principal stochastic-DSA antecedents. This approach, and its descendants in the IMF's Debt Sustainability Framework (IMF, 2013) and Sovereign Risk and Debt Sustainability Framework (IMF, 2022), is the direct ancestor of the Monte Carlo design used here. Closer to the present, Levy Yeyati and Sturzenegger (2023) reformulate the same underlying question as a balance-sheet solvency exercise rather than a flow-based fan chart, valuing a government's assets and liabilities directly; the two approaches are complementary rather than competing, and a balance-sheet cross-check of this paper's own debt projections is a natural extension we have not undertaken.

More recent work has moved from fan charts around a single mean path towards full conditional-distribution estimation. Prasad et al. (2019) introduce growth-at-risk quantile methodology into IMF country surveillance; Lian (Presbitero and Wiriadinata) extend it to the interest-growth differential itself, and Furceri et al. (2025) extend it further into an explicit "debt-at-risk" quantile framework. This is the natural next step for a model such as this one, which currently reports percentiles of a resampled distribution rather than a true conditional quantile regression (a limitation Section 9 returns to). Kaffo Melou (Sumlinski and Geiregat) show both the portability and the limits of the fan-chart approach by applying it outside its original emerging-market sample, to post-HIPC low-income countries, a useful caution against assuming any stochastic-DSA machinery transfers mechanically across very different debt and institutional contexts. Our implementation differs from all of the above by sampling struc-

tural parameters (multipliers, elasticities, growth-accounting inputs) directly from triangular or uniform ranges drawn from named studies, rather than estimating a reduced-form shock covariance matrix; this trades statistical efficiency for interpretability, since every sampled draw can be traced back to a specific coefficient and its source. Bohn (1998) provides the foundational fiscal-reaction-function literature against which any debt-sustainability exercise should eventually be checked. Section A.11 reports what that check looks like for Brazil specifically, and Appendix B reports the diagnostic this model now runs against it.

A.4 Debt thresholds, nonlinearity and the interest-growth differential

Reinhart and Rogoff (2010) and its contested 90%-of-GDP growth-threshold claim, together with the spreadsheet-error and sample-selection correction of Herndon (Ash and Pollin), is the cautionary precedent this paper has in mind whenever it is tempted to summarise a debt path with a single threshold rather than a distribution. Égert (2015) shows why directly: the apparent threshold, where it is detectable at all, is highly sensitive to country and period coverage and typically falls well below 90% once outliers and reverse causality are handled. Pescatori (Sandri and Simon) find no evidence of any specific debt level beyond which medium-term growth collapses, arguing instead that the debt *trajectory*, rising or falling, matters more than the level; Panizza and Presbitero (2014) go further still, using an instrumental-variables strategy built on valuation effects to argue that the negative debt–growth correlation largely disappears once reverse causality is properly addressed, so that no robust causal effect of debt on growth survives. Chudik et al. (2017) reach a related conclusion using dynamic heterogeneous-panel methods across forty countries: no single threshold applies across countries, but a high-debt country on a falling trajectory can grow as fast as a lower-debt peer. Woo and Kumar (2015), by contrast, find a more conventional, if modest, negative panel association operating mainly through slower productivity growth. For Brazil specifically, the only threshold-autoregressive estimate of a Brazilian fiscal reaction trigger we are aware of is Simonassi and Lima (2005), who find that stabilisation is activated only once the deficit crosses an estimated threshold near 1.7% of GDP. That is a different quantity from the debt-level thresholds discussed above, but it is the closest Brazil-specific evidence that fiscal responses themselves may be nonlinear, and it bears directly on the nonlinear risk-premium diagnostic introduced in Appendix B.

A separate literature, arguably more consequential for a country with Brazil's interest-growth gap, asks not whether a debt *level* is dangerous but whether the interest-growth differential itself is structurally favourable or unfavourable. In his 2019 AEA presidential address, Blanchard (2019) argues that historically $r < g$ has been closer to the norm than the exception for the United States, so that debt rollover under such conditions carries limited welfare cost. Furman and Summers (2020) propose real interest payments relative to GDP, rather than the debt ratio itself, as the more informative fiscal-space metric, precisely because it is this differential, not the stock, that drives accumulation; Rachel and Summers (2019) trace the decline in neutral real rates to deeper secular-stagnation forces. Mian (Straub and Sufi,S) formalise how indebted demand depresses the natural rate and, in a model with a zero lower bound, can make the relationship between the interest-growth gap and sustainable fiscal space considerably more complicated than a simple threshold; we take this as a defensible extrapolation of their secular-stagnation argument rather than a result they estimate directly. This literature directly motivates the finding, reported in Section 8, that the interest-growth differential rather than the primary balance is the dominant source of Brazil's own projected status-quo debt accumulation.

A.5 Distributional and social consequences of adjustment

Ball et al. (2013) use a sample of advanced-economy consolidation episodes identified through the Devries et al. narrative approach to show that fiscal consolidation raises the Gini coefficient, lowers the wage share and raises long-term unemployment, with effects that persist for several years after the initial shock. Woo (Bova) extend this to a wider advanced- and emerging-market panel plus country case studies, confirming that spending-based consolidations tend to be more regressive than tax-based ones, though progressive design (protecting targeted transfers, as the present model's default package composition does) can materially offset the distributional cost. Agnello and Sousa (2014, 2012) reach a broadly similar conclusion for eighteen industrialised economies over three decades; Mulas-Granados (2005, 2006) is the earlier European-adjustment-episode precedent on which much of this literature, including Ball et al. (2013), draws for its policy-trade-off framing. More recent work refines the dynamics. Using local-projection methods across seventeen OECD countries, Heimberger (2020) shows the Gini effect builds from around 0.4 points on impact to 0.6 points at medium horizons and is larger for spending-based and post-crisis consolidations, a time profile qualitatively similar to the one our own household microsimulation produces mechanically. Brinca et al. (2021) offer a useful complement running in the opposite causal direction, showing that *higher* initial income inequality itself amplifies the output cost of a given consolidation, a channel this model does not yet capture since its inequality-to-growth feedback (Section 4) remains switched off pending a wired Gini path. Leventi and Matsaganis (2013) use the EUROMOD microsimulation model to separate the distributional effects of fiscal consolidation from those of the accompanying recession in Greece, a methodological strategy close in spirit to this paper's own comparative-statics design. A complementary political-economy perspective comes from Lago-Peñas (Cadaval-Sampedro and Herrero-Alcalde), who show in a survey of Spanish voters that tolerance for austerity depends sharply on its financing instrument and the respondent's own ideology, a reminder that the distributional incidence this model computes mechanically is also, separately, a political variable the model does not attempt to capture. The elasticity our household microsimulation recovers for a comparable Brazilian package, a 0.9-point Gini increase at year eight for an approximately 1% of GDP consolidation, is of the same order of magnitude as this literature's central estimates, though it is produced mechanically from PNAD cash components and a labour-market pass-through rather than estimated from cross-country consolidation episodes, so the agreement is suggestive rather than a validation.

On the health side, Szklo (Clarke and Rocha) estimate the elasticity of infant mortality to public health spending around Brazil's EC 29/2000 constitutional amendment using a difference-in-differences design across municipalities, finding a central elasticity of -0.126, while Gallet and Doucouliagos (2017) provide a meta-regression elasticity of life expectancy to total health spending of 0.04. These Brazilian and cross-country estimates should be read against a standing scepticism: Filmer and Pritchett (1999) find that cross-national public health spending alone explains remarkably little of the variation in child mortality once income and female education are controlled for, which is why this model applies its elasticity only to Brazil's own within-country variation rather than extrapolating a cross-country average. Bokhari (Gai and Gottret) and Moreno-Serra and Smith (2012, 2015) find the mortality-reducing association of public health spending and coverage expansion is in fact stronger in developing- than in advanced-economy samples, at least broadly consistent with a Brazilian elasticity of the magnitude Szklo (Clarke and Rocha) estimate. The strongest Brazil-specific evidence that a targeted transfer programme causally reduces child mortality comes from Rasella et al. (2013), whose study of Bolsa Família finds effects concentrated in poverty- and malnutrition-related deaths, a useful complement to this model's health-spending channel since it operates through a different, income-side mechanism the household microsimulation only partially cap-

tures. Litschig and Morrison (2013); Litschig (2012) identify the effect of municipal spending on schooling outcomes in Brazil using a close-election and a royalty-windfall design respectively, and are the source of this model's schooling elasticity. Hanushek and Woessmann (2012) provide the leading causal, immigrant difference-in-differences evidence that schooling *quality*, not years of schooling alone, drives the human-capital-to-growth channel; Card (1999) remains the standard reference for the likely downward bias of ordinary-least-squares returns-to-schooling estimates relative to instrumental-variables ones. Finally, Acemoglu and Johnson (2007) and Bloom (Canning and Sevilla) offer opposing readings of the health-to-growth channel, the former finding a limited long-run output effect from large historical mortality declines and the latter a sizeable one from cross-country panel estimates, which is why the present model samples this coefficient from a wide uniform range rather than adopting either point estimate.

A.6 Household microsimulation and fiscal-incidence methodology

The household-microsimulation block's comparative-statics design follows a long methodological tradition. Bourguignon and Spadaro (2006) set out the case for static microsimulation as a transparent, auditable tool for evaluating redistribution policy, precisely because every household's exposure to a policy change can be traced, at the cost of foregoing any behavioural response; Immervoll and O'Donoghue (2009) and Figari (Paulus and Sutherland) extend this into the design and validation lessons of large multi-country tax-benefit microsimulation models such as EUROMOD (Sutherland and Figari, 2013). The closest methodological relative to this paper's distributional block among Latin-American-focused work is the Commitment to Equity (CEQ) project, summarised in Lustig (2018)'s handbook and applied to Brazil directly by Higgins and Pereira (2014), who find that regressive consumption taxation offsets much of the redistributive gain from Brazil's direct taxes and transfers, a finding this model's own instrument-specific incidence variants (Section 4) are built to let a user reproduce or contest rather than assume away. Higgins and Lustig (2016) introduce the complementary concept of "fiscal impoverishment": a progressive, inequality-reducing tax-and-transfer system can still leave some households worse off in absolute terms, a distinction this model's poverty-headcount outputs do not yet report separately from its Gini and top-share outputs, a gap worth closing in future work. Lustig (Pessino and Scott) frame the wider Latin American CEQ comparison within which the Brazilian application sits, and Bourguignon (Ferreira and Lustig) is the earlier World Bank volume applying microsimulation and decomposition methods to income-distribution dynamics across Brazil and its neighbours. Closest to the present in method, Balladares and García-Miralles (2026) use administrative tax-record microsimulation to quantify fiscal drag in Spain, a recent demonstration that static, auditable microsimulation of exactly this kind remains a live and publishable way to answer a current policy question, which is the standard this paper's own household block aspires to.

A.7 Sovereign risk premia and fiscal fatigue

The model's debt-dependent risk-premium channel (Section 4) draws its central coefficient from Baldacci and Kumar (2010), but the wider literature on how debt and fiscal fundamentals feed through to sovereign borrowing costs is considerably richer. Edwards (1984, 1986) provide the foundational empirical link between debt and debt-service ratios and the spread charged on developing-country borrowing. A similar relationship holds for a modern panel using EMBI-type spreads: Akitoby and Stratmann (2008) find debt-financed spending raises spreads more than tax-financed spending of the same size, a composition effect this model's revenue-elasticity channel (Section 4) is built to let the user explore. Ardagna (Caselli and Lane) find the debt

effect on long-term rates in OECD economies is concentrated in periods when debt is *above* its own historical average, an empirical analogue, well short of a structural model, of a nonlinear or threshold-dependent premium. The explicitly nonlinear theoretical case is made most directly by two papers: Ghosh et al. (2013)'s "fiscal fatigue" framework models a primary-balance reaction function that flattens as debt rises, producing an endogenous fiscal limit, while Bi (2012) derives a stochastic fiscal limit from a dynamic Laffer curve along which risk premia rise nonlinearly once debt passes a country-specific threshold. These two papers are the direct theoretical basis for the nonlinear, fiscal-fatigue-style risk-premium diagnostic introduced in Appendix B. Reinhart (Rogoff and Savastano)'s concept of "debt intolerance", the finding that safe debt thresholds are far lower, and more history-dependent, for emerging markets than for advanced economies, further motivates treating any single premium coefficient as provisional rather than structural; Mendoza and Yue (2012) provide the leading general-equilibrium unification of sovereign default and the business cycle, a theoretical benchmark against which this model's reduced-form premium channel is a deliberately modest approximation. For Brazil and its emerging-market peers specifically, Gupta (Mati and Baldacci) find fiscal consolidation narrows spreads especially for countries with a default history, and Campos and Cysne (2021) estimate country-specific rather than common debt limits for eighteen emerging economies, rejecting a single threshold in a manner consistent with Égert (2015)'s growth-threshold finding above. Ornelas (2017) and Rocha and Moreira (2012), both Brazilian-institution-authored, provide the closest direct estimates of the debt-to-spread elasticity for Brazil and comparable emerging markets, broadly consistent with, if somewhat larger than, the Baldacci and Kumar (2010) coefficient this model currently uses as its central case.

A.8 Governance, corruption and public-spending leakage

A distinct but related literature asks how much of public spending is diverted before it reaches its intended beneficiaries, and how this interacts with fiscal policy. Ferraz (Finan and Moreira) use random federal audits of Brazilian municipal education transfers to estimate that corruption is found in roughly a third of audited municipalities, diverting around 8% of audited education resources on average. Using the same audit programme, Avis (Ferraz and Finan) show that a past audit visit reduces subsequently measured corruption by about 8% across nearly two thousand municipalities, evidence this model uses as a lower bound on how much a credible anti-corruption programme could remove. Litschig and Zamboni (2018) extend this audit literature with a randomised increase in audit risk across Brazilian municipalities and find procurement corruption falls by 10–15 percentage points, though health- and cash-transfer-programme compliance is unaffected, a useful caution that the leakage parameter's effectiveness may be programme-specific rather than uniform (a simplification the present model's single leakage share does not yet relax). Colonnelli and Prem (2022) and the companion firm/worker-level study of Colonnelli et al. (2022) use the same Brazilian federal audit programme, the Controladoria-Geral da União (CGU), to show audits raise firm entry and growth in government-dependent sectors while penalising politically connected firms, rejecting a "grease the wheels" reading of corruption. Using discontinuities in Brazilian federal transfers, Brollo et al. (2013) show that larger unearned fiscal windfalls themselves raise measured municipal corruption and lower the quality of elected candidates, a result directly relevant to any scenario, such as a commodity-revenue boom, in which this model's external block delivers an unusually large fiscal windfall. Bobonis (Cámara Fuertes and Schwabe) provide a contrasting Puerto Rican result: pre-election audits reduce corruption and raise reelection chances, but the deterrent effect does not persist into later, non-electoral audit rounds, a caution against assuming the Brazilian audit evidence this model relies on implies a permanent, rather than

temporary, leakage reduction.

Mauro (1995) established the cross-country association between perceived corruption and lower investment and growth that motivates treating corruption as a growth-relevant variable at all. Mauro (1998) extends this, showing corruption systematically skews public spending away from education and towards more easily diverted categories, a composition effect related to, but distinct from, the leakage-share mechanism used here; Tanzi and Davoodi (1997) similarly find corruption raises the *volume* of public investment while lowering its productivity and the revenue available to finance it, a quantity-versus-quality distinction this model's single leakage parameter does not yet separate. Gründler and Potrafke (2019) provide a more recent panel estimate of the long-run output effect of an improvement in the Transparency International Corruption Perceptions Index, which we hold off by default pending verification of a cross-country standard-deviation input. Hanousek and Kočenda (2011) close the loop most directly for this paper's purposes, linking corruption and regulatory quality to public investment, deficits and debt across the new EU member states, and finding that lower corruption is associated with better fiscal outcomes on both margins, a structural pairing of governance and debt dynamics that this model keeps as two formally separate blocks (G1/G2 and the debt identity) rather than one estimated relationship. The benchmark for the difficulty of measuring leakage directly, rather than through perceptions, is Olken (2007)'s missing-expenditure approach on Indonesian road projects. A closely related finding is Reinikka and Svensson (2004)'s "local capture" result, that only around 13% of an Ugandan school capitation grant reached its intended schools on average, and Fisman and Svensson (2007), who show using Ugandan firm-level data that bribery has roughly three times the negative effect on firm growth of the statutory tax rate, supporting a "corruption as tax" framing. Golden and Picci (2005) propose inferring corruption objectively from the gap between reported infrastructure spending and the physical infrastructure actually delivered, a methodological contrast to the survey-based perception indices this model explicitly avoids using as a model input (Section 4). Our leakage parameter is calibrated on the comparable Brazilian audit evidence rather than attempting a similar missing-expenditure exercise, which the available Brazilian audit microdata does not support at the scale required.

A.9 Fiscal rules and the political economy of public debt

Both 2026 candidate programmes modelled in Section 5 are, formally, fiscal rules (an expenditure rule and a balance-targeting rule), and so inherit a large, separate literature on fiscal-rule design that the model's own machinery is agnostic to but that any reader evaluating the two programmes' relative merits should have in view. Kopits and Symansky (1998) provide the foundational IMF taxonomy of rule types, balanced-budget, debt, expenditure and revenue rules, and the credibility-versus-flexibility trade-off between them. Using a dataset spanning eighty-one countries, Schaechter et al. (2012) document that post-2008 rules increasingly combine several rule types with escape clauses, a description that fits Brazil's own layered combination of the constitutional spending ceiling (Complementary Law 200/2023) and the pre-existing "golden rule" (Article 167, assessed directly for Brazil by Pires, 2019) reasonably well. Cordes et al. (2015) is the closest empirical comparator to the incumbent's programme specifically, evaluating the compliance record and investment effects of thirty-three expenditure rules worldwide. The standard real-world evaluation of an operating expenditure-ceiling-type rule is Feld and Kirchgässner (2008)'s study of the Swiss debt brake, which finds it durably reduced deficits without the escape-clause erosion some critics predicted, a useful, if institutionally distant, comparator for assessing whether Brazil's own ceiling is likely to bind. López-Villavicencio and Zoumenou (2025) offer a timely, directly comparable recent finding for the

European Union, that only compliance with a binding deficit or debt rule, not an expenditure rule considered alone, robustly improves subsequent fiscal outcomes, which is worth weighing against the Brazilian ceiling's own design. Debrun and Kumar (2007) and Wyplosz (2018) examine the complementary role of independent fiscal councils in making numerical rules credible, relevant to Brazil's own Independent Fiscal Institution (IFI), whose most recent technical assessment of the 2023 spending framework we draw on directly in Section 5 (Instituição Fiscal Independente, 2023); Debrun (Hauner and Kumar) survey the wider international evidence on independent fiscal agencies. At a more foundational level, Alesina and Tabellini (1990) and Alesina and Passalacqua (2016)'s handbook survey model why governments with divergent political preferences over-borrow relative to a social optimum, a channel that operates independently of whichever numerical rule is nominally in force, while von Hagen and Harden (1995) and Persson and Tabellini (1999) are the classic references connecting budget-process centralisation and electoral-system design to fiscal discipline more broadly. None of this literature is wired into the model; it is offered here as the frame within which a reader should situate the two 2026 programmes' chosen closure rules, which this paper deliberately declines to rank (Section 5).

A.10 Stock-flow adjustments, quasi-fiscal operations and fiscal transparency

The model's single most consequential finding is that its debt-level back-test fails specifically because of discretionary, non-forecastable "stock-flow adjustment" operations (Section 7), and this sits within a literature that has independently documented the same phenomenon across many countries. Using 163 countries over three decades, Weber (2012) shows stock-flow adjustments are a large and persistent, not merely occasional, source of debt accumulation worldwide, only partly explained by observable balance-sheet effects and contingent liabilities, and systematically smaller in more fiscally transparent countries, directly consistent with this paper's own finding that Brazil's residual bears no relationship to any observable macroeconomic covariate. Seiferling (2013) refines this using government balance-sheet data directly, arguing that complete financial reporting, not a transparency *score* per se, is what narrows the gap. The closest empirical analogue to this paper's own 3.9-percentage-point single-year residual is Jaramillo (Mulas-Granados and Kimani)'s examination of 179 historical debt-spike episodes: spikes are driven primarily by stock-flow adjustments rather than by primary deficits or weak growth, and a large stock-flow adjustment predicts debt subsequently *failing* to decline, a genuinely alarming precedent for interpreting Brazil's own recent experience. Andrian (Rodríguez and Valencia) extend this to a hazard-model framework across 172 countries, finding a one-percentage-point rise in the stock-flow adjustment raises the probability of a debt surge by roughly fifteen per cent. Afonso and Jalles (2019) show that stronger fiscal-rule frameworks moderate the stock-flow-adjustment channel, while Casalin (Cerniglia and Dia) show that in many countries stock-flow adjustments function as a hidden, debt-management-driven component of the true cost of debt service, a framing that fits the Brazilian Treasury-to-BNDES lending and Central Bank result-transfer operations this paper identifies as the dominant residual unusually well. The foundational IMF treatment of *why* such operations occur at all is Mackenzie and Stella (1996): public financial institutions and central banks routinely run "quasi-fiscal" activities that shift genuine fiscal costs off the conventional budget, a description that applies directly to BNDES's role in the Brazilian episode. Bova et al. (2016) assemble the first systematic cross-country dataset of realised contingent-liability costs, finding an average fiscal cost of around six per cent of GDP, far higher for financial-sector bailouts, that clusters in crises and is, like this paper's own SFA residual, poorly predicted by observable covariates. Alesina (Prati and Tabellini) provide the classic theoretical link between debt-management

choices, such as maturity structure and market confidence, and self-fulfilling crisis risk. Finally, the European Commission's own recurring statistical reporting on member-state stock-flow adjustments (Eurostat, 2024) documents magnitudes of a similar order, up to two to three per cent of GDP, even among countries operating under a far more codified fiscal-transparency regime than Brazil's, suggesting the phenomenon this paper documents is a structural feature of sovereign debt accounting rather than a peculiarly Brazilian failing. No peer-reviewed economics-journal article appears to model Brazil's own 2015–2016 *"pedaladas fiscais"* episode specifically as a stock-flow-adjustment event; the closest institutional treatment, which we draw on directly instead, is Gobetti and Orair (2017)'s reconstruction of Brazil's above-the-line fiscal accounts net of the same operations (Section A.11).

A.11 Brazilian macro-fiscal evidence

Sanches (2026) update Brazilian federal fiscal multiplier estimates by spending and revenue instrument using a structural approach over federal accounts from 1997 to 2023, and Carvalho and Sanches (2019) is the antecedent study on which the present model's package-specific multipliers draw. Gruss (2014) estimates how strongly Latin American and Caribbean output responds to commodity terms-of-trade shocks and finds, in an input this model inherits directly, that Brazil's response is markedly smaller than its regional peers; our own back-test (Section 7) suggests this may understate the 2003–2010 commodity-boom experience once capital inflows and valuation effects are considered. Bom and Ligthart (2014) provide the meta-analytic range for the output elasticity of public capital that this model uses, pending the Brazilian public-capital stock data needed to activate that channel, and Ostry (Berg and Tsangarides) is the IMF staff discussion note on redistribution, inequality and growth that motivates an inequality-to-growth feedback the model specifies but, pending a wired Gini path, does not yet activate.

Two further bodies of Brazil-specific evidence bear directly on this paper's own findings. First, on the fiscal reaction function Section A.3 flagged as the natural check on any debt-sustainability exercise: de Mello (2008) estimates a conventional, Bohn-style reaction function for Brazil across three levels of government (1995–2004) and finds the primary surplus responds positively and significantly to debt, in line with the typical emerging-market panel result; Mendonça (Santos and Sachsida) refine this with a Markov-switching specification and find a structural break around 2000, with a strong reaction pre-2000 and a weak-to-absent one thereafter. Both estimates are of the *conventional* sign, the opposite of the small, negative Brazilian-specific coefficient this paper's own D3 debt-panel review finds over 2000–2024 (Appendix B), a discrepancy we read not as a contradiction but as evidence that Brazil's fiscal reaction has itself shifted, plausibly again, since the mid-2000s sample these two papers cover, an open question this paper does not resolve. Second, on the stock-flow-adjustment residual at the centre of this paper's own back-test failure: Gobetti and Orair (2017) reconstruct Brazil's above-the-line fiscal accounts net of the same discretionary operations (Treasury-to-BNDES lending, Central Bank result transfers, and the broader *pedaladas fiscais* episode) that this paper's own residual isolates, the closest peer-reviewed treatment of that phenomenon available; Orair and Gobetti (2017) provide the companion narrative of Brazilian fiscal policy's swing from 2005–2010 expansion through 2011–2014 tax cuts to post-2015 austerity, useful context for interpreting why the residual is concentrated in that period. Bevilaqua and Garcia (2002) and Vivian and Mendonça (2008) cover Brazilian public-debt management strategy more broadly; Nunes et al. (2020) show how Central Bank profit transfers have themselves been used to help satisfy the constitutional golden rule even as public investment has lagged debt growth, a fiscal-rule-gaming concern directly relevant to Section A.9; and the Brazilian Senate's own Independent Fiscal Institution provides the most current technical evaluation of the 2023

spending framework modelled in Section 5, flagging its reliance on uncertain, court-dependent revenue (Instituição Fiscal Independente, 2023), a companion technical note on integrating the golden rule and the spending ceiling into one coherent architecture (ENAP / Tesouro Nacional, 2019). Finally, Alston et al. (2013) frame Brazil's post-1985 fiscal and institutional trajectory as a slow change in beliefs about the social contract itself, a political-economy lens worth bearing in mind whenever two candidate programmes are described, as this paper describes them, as formally interchangeable closure rules for the same debt-accumulation identity.

B Diagnostic extensions: how much would a reaction, or a nonlinear premium, change the picture?

This appendix gives the full detail summarised in Section 7.1. Tier 2's failure (Section 7) leaves open a question the back-test itself cannot answer: how much of the headline status-quo debt path is an artefact of the model's deliberately simple, no-policy-reaction specification, and how much would a more behaviourally realistic government, or a more aggressively nonlinear market response, change it? Rather than fold a speculative answer into the headline results, we report it as a clearly labelled, off-by-default diagnostic, run only against the status-quo baseline (S0) itself, never differenced against or compared with the policy scenarios of Section 5, and excluded from every published Monte Carlo summary. Three hundred draws (central draw plus 299 sampled, seed 20261020, an RNG stream independent of every other Monte Carlo channel) are used throughout. Every absolute debt figure in this appendix, breach probabilities, debt levels, the fiscal gap, inherits the same Tier 2 bias documented in Section 7: the model's own seven-year recursion overshoots observed debt by 12.5–16.1 pp of GDP even after the best available correction, and nothing here adjusts for that overshoot. We flag this once, at the top, rather than repeat it after every number, but it applies with equal force throughout, including to the breach probabilities below, which should be read as upper bounds conditional on an uncorrected model rather than as calibrated risk estimates.

First, an EC/IFI-style probability-of-breach and fiscal-gap statement replaces the bare median debt path with a distribution. Table 2 reports the probability that S0's debt ratio exceeds 90%, 100% and 120% of GDP at each horizon, with and without the stochastic stock-flow-adjustment term of Section 7; Table 3 reports the primary balance that would stabilise debt at each horizon's own implied interest-growth differential (in the spirit of the European Commission's S1/S2 fiscal-gap indicators), against S0's actual projected primary balance. One feature of Table 2 is worth flagging rather than smoothing over: adding the (demeaned) stock-flow-adjustment term slightly *lowers* the five-year breach probability (78.6% to 73.6%) before the usual pattern of added uncertainty raising breach probabilities reasserts itself at longer horizons. We attribute this to the interaction between the bootstrapped adjustment and the debt-dependent risk-premium feedback channel of Section 4, which can make a mean-preserving spread shift probability mass in either direction around a fixed threshold depending on where the threshold sits in the resulting distribution; we have not fully decomposed this interaction and report the counter-intuitive short-horizon result rather than suppress it.

Table 2: Probability that S0 debt/GDP exceeds stated thresholds, by horizon (300 draws).

Horizon	$P(>90\%)$, no SFA	$P(>90\%)$, SFA	$P(>100\%)$, SFA	$P(>120\%)$, SFA
5 years	78.6%	73.6%	35.8%	0.7%
10 years	97.7%	95.0%	86.3%	61.5%
15 years	100.0%	99.7%	98.0%	89.0%
20 years	100.0%	100.0%	100.0%	98.3%

Table 3: Stabilising primary balance and fiscal gap, S0, pp of GDP. Median [p10, p90] across 300 draws.

Horizon	Stabilising primary balance	Actual projected primary balance	Fiscal gap
5 years	+2.60 [−0.22, +4.84]	−2.39 [−2.44, −2.22]	+4.85 [+2.11, +7.16]
10 years	+3.60 [−0.14, +7.16]	−4.04 [−4.17, −3.68]	+7.42 [+3.74, +11.13]
20 years	+7.95 [+0.65, +17.26]	−7.68 [−8.45, −6.80]	+15.63 [+8.06, +25.10]

Second, we report two off-the-side variants of S0 itself, holding its own interest-growth path fixed and changing only the primary-balance rule or the risk-premium shape, so that each isolates a single mechanism. `frf_emerging` recomputes the primary balance each year as $pb_t = pb_{S0,t} + \rho\,(d_{t-1} - d_{2025})$ using the typical emerging-market panel's estimated long-run fiscal-reaction coefficient ($\rho = 0.0334$). Brazil's own estimated coefficient over 2000–2024, an in-house estimate produced for this project's own debt-panel review, not an independently peer-reviewed figure, and reported here with that provenance made explicit, is small and of the *opposite* sign (−0.037), in contrast to the conventional, positive, peer-reviewed coefficients de Mello (2008) and Mendonça (Santos and Sachsida) estimate for Brazil over earlier, pre-2005 samples (Section A.11); we therefore also report `frf_brazil_robustness`, which applies this in-house coefficient instead, read as a robustness check on an internal estimate rather than as a published external finding. `premium_nonlinear` replaces the linear, capped risk-premium channel of Section 4 with the fiscal-fatigue-style shape theorised by Ghosh et al. (2013) and Bi (2012) (Section A.7): zero below a historical threshold debt level (119.6% of GDP, the 95th percentile of a fifty-country comparator panel) and rising with the square of the excess above it, capped at 2,000 basis points to avoid the numerical explosion an uncapped quadratic specification produces. Table 4 reports all three alongside plain S0, where "plain S0" is the *median across the 299 sampled draws*, not the single deterministic central-parameter projection reported in Sections 4 and 8 (145.5% of GDP at this horizon). The two are different statistics and need not coincide: the median of a resampled, compounding debt-path distribution is not generally equal to the same path evaluated at the sampled parameters' own central values, since the debt identity compounds multiplicatively and the sampled distribution is right-skewed. We report both, rather than letting one stand in for the other, precisely because conflating them would overstate how tightly the two exercises agree.

Table 4: S0 debt/GDP (%) under diagnostic reaction and premium variants. Median [p10, p90] across 299 sampled draws throughout, including the "plain S0" column (distinct from the 145.5%-at-ten-years deterministic central projection of Table 1's own baseline; see text).

Horizon	Plain S0 (sampled median)	`frf_emerging`	`frf_brazil_robustness`	`premium_nonlinear`
10 years	130.4	124.0 [99.3, 145.6]	138.9 [104.8, 168.8]	130.5 [101.7, 157.1]
20 years	245.5	206.7 [146.4, 270.9]	308.8 [184.8, 441.8]	487.8 [165.8, 1080.0]

Three readings of these tables deserve emphasis, each a direct application of the literature surveyed in Sections A.3–A.11. First, under the model's own preferred specification (including the stochastic stock-flow-adjustment term), the probability of breaching even a comparatively undemanding 90% of GDP threshold is already 73.6% at five years and 95.0% by ten (Table 2's "SFA" columns, not the more optimistic "no SFA" columns), a statement Table 1's paired differences cannot make on their own, and precisely the kind of risk statement Garcia and Rigobon (2004) and the growth-at-risk literature (Prasad et al., 2019; Lian, Presbitero and Wiriadinata; Furceri et al., 2025) argue a central path alone obscures. Second, the `frf_brazil_robustness`

variant, using Brazil's own, historically estimated reaction coefficient rather than the emerging-market average, produces a *higher*, not lower, debt path than plain S0, the arithmetic consequence of that coefficient's counter-intuitive sign; we are careful to present this only as a robustness check on what Brazil's own history implies, following Simonassi and Lima (2005)'s related finding that Brazilian fiscal reaction may itself be regime-dependent, and explicitly not as a forecast that Brazilian fiscal policy will behave this way going forward. Third, the nonlinear premium variant is flat and close to the linear specification below the historical threshold but diverges sharply above it, reaching a 90th-percentile debt ratio of 1,080% of GDP at twenty years, a number we report not as a plausible outcome but as a crisis-territory flag in the sense Ghosh et al. (2013) and Bi (2012) intend: once a fiscal-fatigue-style premium is allowed to operate, paths that cross the historical threshold do not approach a new equilibrium, they explode, which is itself the theoretical point. None of the three variants alters any other published Monte Carlo output, and none is a forecast: each isolates a single mechanism against an otherwise unchanged baseline, in the same spirit as the channel-by-channel attribution already used for the long-run feedback block (Section 4).

C Diagnostic extensions, part II: responding to an independent review

An independent review of an earlier version of this paper (summarised in Section 9) argued that the model is acyclical, monetary-policy-blind and currency-blind, and that this quietly biases the finding that expansion scenarios lower the Gini coefficient (Table 1) in favour of expansion, because the model never prices the channels a sustained, debt-financed expansion would plausibly trigger: a Selic hike, a weaker real, and interest income accruing disproportionately to bondholders concentrated at the top of the income distribution. This appendix reports, in full, six off-by-default diagnostics built in direct response, each isolating one mechanism against the unchanged baseline, in the same spirit as Appendix B. None is folded into Table 1 or any other published Monte Carlo summary; each uses 300 draws (central draw plus 299 sampled, seed 20261026, independent RNG streams) unless stated otherwise. Section C.5 adds a seventh, different in kind from the other six: not a Monte Carlo diagnostic at all, but a direct test of whether MR1's reactive-Selic mechanism, used as a stand-in for the model's own fixed-rate assumption, actually improves the fit to Brazil's observed fiscal history.

C.1 Does the model consider the business cycle?

The review's starting diagnosis was correct, and independently confirmed twice over. First, this model's own deterministic growth trend has no cyclical fluctuation whatsoever: under the central draw with no external shocks, S0's year-on-year real GDP growth decelerates smoothly and monotonically from 1.18% in 2026 to 0.94% by 2035, with no booms or slowdowns at all. Second, a historical fiscal-timing diagnostic against Brazil's own 2000–2024 record (gross domestic product growth against its own trailing five-year mean, and the year-on-year change in the primary balance) finds fiscal policy was predominantly *countercyclical* over the period, not procyclical: 2009 and 2020–2021 are both clearly countercyclical, and only two years in the whole sample, 2006 and 2023, classify as procyclical-expansionary. The review's own singled-out 2011–2014 period is mixed rather than uniformly procyclical: 2011 is procyclical-contractionary, but 2012–2014 are all countercyclical on this measure, a finding that partly complicates, rather than confirms, the review's qualitative reading of that episode.

Reintroducing genuine cyclical variation via the model's existing stochastic external-shock bootstrap (Section 4) and rescaling the fiscal multiplier by a sampled recession/expansion ratio (low 1.0, central 1.5, high 2.5, reflecting Auerbach and Gorodnichenko (2012)'s qualitative finding of larger multipliers in recessions and Ramey (2019); Ramey and Zubairy (2018)'s

more sceptical reading of the same literature; Pires (2014), cited elsewhere in this paper for Brazilian multiplier estimation generally, is not used for this specific ratio, since a secondary source suggests its own positive finding concerns a volatility-regime split rather than a business-cycle-phase split) moves R_AMB's median ten-year GDP cost from −0.80% under the model's fixed multiplier to −1.08% once state-dependence is reintroduced, roughly 35% larger, consistent with the review's intuition that ignoring state-dependence understates the true cost of a sustained contraction.

C.2 Debt composition and the exchange rate

The debt identity (Section 4) applies one interest rate to an undifferentiated stock and revalues foreign-currency-linked debt only indirectly, through inflation pass-through, never as a direct stock revaluation. Section 9 already reports why this channel is structurally small for Brazil specifically (a stable 3.7–4.4% foreign-exchange-linked share of federal debt, Secretaria do Tesouro Nacional, RMD); this appendix quantifies it. Table 5 revalues that FX-linked share directly by the model's own bootstrapped BRL/USD path (common to every scenario, as in Section 4). The effect is indeed small at the paper's main ten-year horizon, as Section 9's framing expects, but grows to a no-longer-negligible median 5–17 pp of GDP by twenty years, because the small FX-linked tranche is revalued by *cumulative* depreciation, and Brazil's own 2000–2024 BRL/USD history includes twenty-year windows with well over 100% cumulative depreciation; this remains modest relative to the already very wide uncertainty band on total debt/GDP at that horizon, but the "second order, do not worry about it" framing should not be extended to the twenty-year horizon without this qualification.

Table 5: FX-linked debt revaluation: additional debt/GDP (pp) from revaluing the FX-linked tranche by the common bootstrapped BRL/USD path. Median [p10, p90], 300 draws.

Scenario	10 years	20 years
S0	+3.21 [−0.48, +10.39]	+16.93 [+1.84, +47.50]
R_MOD	+2.92 [−0.43, +9.44]	+13.46 [+1.38, +37.54]
R_AMB	+2.63 [−0.39, +8.46]	+9.79 [+0.94, +28.46]

Separately, FX1 asks whether the exchange rate should itself respond to a scenario's own fiscal stress, rather than only to the common, policy-blind path above. Rather than fit a new debt-to-FX regression — this paper's own external-sector block already tried the closest analogue, a debt-to-sovereign-spread relationship for Brazil (fit 2000–2014, tested 2015–2024), and found it not supported out of sample — FX1 chains the model's own already-estimated debt-to-risk-premium parameter (Baldacci and Kumar, 2010) to a sampled, not fitted, spread-to-depreciation elasticity anchored on uncovered interest parity, with its upper bound informed only qualitatively by Banerjee (Mehrotra and Zampolli)'s finding that the fiscal-to-exchange-rate channel is markedly stronger in emerging than advanced economies. A first implementation routed this extra, stress-induced depreciation through the same inflation-pass-through channel as the common path above and found an *exactly zero* debt/GDP effect at every draw: in this model's Fisher-consistent debt identity, the nominal interest rate and nominal GDP growth both reprice by the same inflation factor, so a pure inflation shock with no real-economy effect cannot move the debt/GDP ratio at all, by construction — an independently confirming structural fact about the model, in the same spirit as the acyclical growth core reported above. FX1 therefore revalues the same FX-linked tranche as Table 5, but by the scenario's own stress-induced depreciation rather than the common path. Table 6 shows this is, on its own, a small effect for the paper's expansion scenarios, confirming that the FX-linked channel stays small

specifically because the underlying FX-linked share of debt is small, not because fiscal stress fails to move the exchange rate.

Table 6: FX1: additional debt/GDP (pp) from a stress-induced, scenario-specific depreciation revaluing the FX-linked tranche. Median [p10, p90], 300 draws.

Scenario	10 years	20 years
E_MOD	+0.18 [+0.06, +0.36]	+0.42 [+0.16, +0.83]
E_AMB	+0.21 [+0.08, +0.40]	+0.47 [+0.18, +0.93]

C.3 Interest income and the inflation tax

The review's most direct distributional claim was that higher government interest expense accrues disproportionately to bondholders at the top of the income distribution, and that an inflation tax falls hardest on poor households' non-indexed transfers — neither of which the household microsimulation (Section 4) had any mechanism to represent, since the underlying survey has no capital-income field. IY1 derives a scenario's change in government interest expense directly from the debt identity (the previous year's debt/GDP times that year's nominal interest rate), allocates it across the simulated population using the concentration of declared capital and financial income among Receita Federal's own tax centiles (Receita Federal, 2024) (the top 1% of declarants hold 68.3% of declared capital/financial income; the top 10% hold 90.7%), and separately erodes each household's baseline Bolsa Família amount by that scenario-year's cumulative extra inflation. Table 7 reports the resulting shift in the Gini coefficient against the published, IY1-off figure, for the two debt-financed expansion scenarios the review's claim is specifically about.

Table 7: IY1: shift in the Gini coefficient (points) from switching on the interest-income and inflation-tax channels, against the published figure. Median [p10, p90], 300 draws.

Scenario	10 years	20 years
E_MOD	+0.76 [+0.41, +1.16]	+3.13 [+1.60, +4.56]
E_AMB	+1.31 [+0.72, +1.98]	+5.02 [+2.48, +7.19]

Poverty (the R$218 line) barely moves anywhere in this table (well under one percentage point throughout), which confirms the capital-income windfall, not the inflation-tax erosion of Bolsa Família, is the dominant channel here: poverty is measured at the bottom of the distribution, and the windfall is concentrated at the top. The direction is exactly what the review predicted, and the magnitude is not negligible against the published figure it is being compared with, which we turn to directly in Section C.6 below.

C.4 A reactive Selic

The review's remaining claim was that the model has no monetary-policy reaction function: the real interest rate is a fixed input regardless of inflation or the output gap a scenario itself creates. Rather than make this shared rate-determination function itself state-dependent, which would touch every scenario, every other diagnostic in this appendix, and the published Monte Carlo at once, MR1 re-runs each scenario's debt path afterwards, substituting a textbook Taylor rule (Taylor, 1993) for the fixed rate in an otherwise identical re-run of the debt identity. Because this model's inflation input does not vary by scenario, only the output-gap term differentiates a scenario's Selic path from S0's. The Brazil-specific Taylor-rule literature we could locate

(several studies spanning 2001–2025) consistently reports a positive, significant Selic response to both the inflation and output gap, but only qualitatively ("smaller than advanced economies," "varying over time"); rather than use an unverified point estimate, the coefficient is sampled as a discount (low 0.3, central 0.6, high 1.0) on Taylor's own classic 0.5 coefficient. Table 8 reports the resulting incremental debt/GDP, on top of the already-reported fixed-rate baseline differential (Table 1).

Table 8: MR1: incremental debt/GDP (pp) from a reactive Selic, on top of the fixed-rate baseline differential vs. S0. Median [p10, p90], 300 draws.

Scenario	10 years	20 years
E_MOD	+1.46 [+0.83, +2.39]	+5.45 [+2.92, +10.26]
E_AMB	+3.02 [+1.71, +4.95]	+6.32 [+3.28, +12.66]

C.5 Testing MR1 against Brazil's own fiscal history

MR1 above answers one question: within a Monte Carlo draw, how would a reactive Selic change a scenario's debt path relative to S0? A different and harder question is whether MR1's mechanism, used not as an add-on to a scenario comparison but as a direct replacement for the model's own fixed real-rate assumption, actually describes Brazil's observed 2000–2024 fiscal history better than the fixed rate it would replace. This is the same distinction Section 7 already draws for the published model's own back-test, Tier 1 (differences) against Tier 2 (absolute levels): a mechanism can be defensible for one purpose and not the other, and the two tests need not agree.

We built a separate historical reconstruction, outside any Monte Carlo draw: starting from Brazil's own observed general-government debt/GDP in 2006, the debt identity is recursed annually through 2024 using the actually observed primary balance and real GDP growth every year, but *generating*, not observing, the interest rate under two candidate mechanisms — the model's existing fixed real rate (5.05%, itself the historical median real rate the model already uses, not an arbitrary flat guess) and MR1's Taylor rule, reacting each year to that year's own observed inflation and a real-time output-gap proxy identical to the one Appendix B's PCY1 diagnostic already uses. Two debt-level tests compare each mechanism's resulting path to the actually observed one: a cumulative reconstruction from the single 2006 anchor, which Section 7 already documents as dominated by stock-flow-adjustment noise unrelated to any rate mechanism, and a one-step reconstruction re-anchored on the observed lagged debt every year, which isolates rate-mechanism fit from that noise and is accordingly the test we treat as decisive. A third test compares each mechanism's own implied nominal rate directly to the actually observed Selic.

Table 9 reports the result, confirmed against two independently constructed nominal-growth series (a CPI-and-real-growth approximation, and Brazil's own SIDRA current-price GDP series, used elsewhere in this paper's growth-block back-test). The two tests disagree, and we report that disagreement plainly rather than resolve it in either direction: the Taylor rule's own implied rate tracks the actually observed Selic more closely than the fixed-rate assumption does, but feeding that same generated rate through the debt identity produces a *worse* one-step reconstruction of the observed debt path than the fixed-rate assumption. We read this as a category error our own design would have invited, not a contradiction: the fixed rate is calibrated to the government's effective, blended rate across its whole diversified debt stock (Selic-linked, inflation-linked and fixed-rate instruments alike, Section 9), not the Selic policy rate itself, and a rule built to track Selic is accordingly the wrong direct stand-in for it. We

therefore do not adopt MR1's mechanism as a replacement for the fixed-rate assumption, in the published Monte Carlo or elsewhere; Table 8's own incremental figures remain exactly what they already were, a Tier-1-style scenario-difference diagnostic, not a claim that the underlying mechanism is a validated improvement on the model's historical fit. Nominal GDP growth in the CPI-based series is a Fisher approximation (real growth compounded with CPI inflation) rather than a directly observed nominal series; this affects both mechanisms' reconstructions equally and so does not bias the comparison between them, which is the comparison this test exists to make.

Table 9: MR1 against history: one-step debt/GDP reconstruction error and nominal-Selic tracking error, fixed-rate vs. Taylor-rule mechanism, 2000–2024 (RMSE, percentage points).

Mechanism	Debt/GDP, one-step (CPI-based)	Debt/GDP, one-step (SIDRA-based)	Nominal Selic
Fixed rate	3.69	3.23	3.87
Taylor rule	4.25	3.88	3.35

C.6 Putting it together: does the bias claim hold up?

The review's claim was that the Gini and debt findings of Table 1 are "probably biased in favour of expansion." Combining each diagnostic's own median effect with the corresponding published figure — a simple addition of two separately-run statistics, not a single re-integrated Monte Carlo, since medians of compounding, nonlinear quantities do not generally add linearly, so this is indicative rather than a recomputed joint distribution — gives a direct answer for the two channels large enough to matter at the paper's main ten-year horizon.

On distribution: the published finding is that E_MOD lowers the Gini coefficient by a median 0.10 points and E_AMB by 0.16 points at ten years (Table 1). Adding IY1's own median shift (Table 7) gives a combined $-0.10 + 0.76 = +0.65$ points for E_MOD and $-0.16 + 1.31 = +1.15$ points for E_AMB: the median effect does not merely shrink, it *reverses sign*. On this reading, the review's claim is not just plausible but directionally confirmed: once interest income is distributed the way Receita Federal's own data says capital income actually is, the model's central estimate is that these expansions make the Gini coefficient worse, not better, at the median. On debt: adding MR1's and FX1's own median effects (Tables 8 and 6) to the published differential raises E_MOD's ten-year debt cost from +11.6 to roughly +13.3 pp of GDP and E_AMB's from +23.3 to roughly +26.5 pp, about 13–14% larger in each case — a real, quantified addition, though one that changes the *size* of the debt cost rather than its sign.

We think this is the honest answer the review's own methodological standard demands, stated plainly rather than hedged: the distributional claim this paper previously reported without these channels does not survive their inclusion at the median, and the debt claim, while directionally unchanged, is understated by a magnitude worth citing rather than waving away. We did not set out to find this; PCY1 and FX1, by contrast, *complicate* rather than confirm parts of the review's framing (fiscal policy's own history is predominantly countercyclical, and the FX-linked debt channel is genuinely small once isolated from the common external path), and CYC1's and DC1's findings are each real but, at the ten-year horizon that is this paper's main reference point, modest. The net reading is neither "the review was simply right" nor "the original paper's finding stands": it is that the specific channel the review pressed hardest on, interest income flowing disproportionately to the already-wealthy, is large enough to reverse a headline distributional finding once it is actually quantified, and that is precisely the kind of result an honest accounting exercise exists to surface, whichever way it comes out.